AMCA: Attention-Guided Multiscale Context Aggregation Network for Remote Sensing Image Change Detection

Xintao Xu, Zhe Yang, and Jinjiang Li

***Abstract*— Remote sensing image change detection is the key to understanding surface changes. Although the existing change detection methods have achieved good results, some structural details are missing and the detection accuracy needs to be improved. Therefore, we propose an attention-guided multiscale context aggregation network (AMCA) for remote sensing image change detection. First, we use the fully attentional pyramid module (FAPM) to enhance the deep feature information of the original image. And we introduce the dense feature fusion module (DFFM) to fully fuse the bitemporal features to obtain the change regions. Second, the introduction of channelwise cross fusion transformer (CCT) and channelwise cross attention (CCA) not only can effectively fuse channel features focusing on different semantic patterns but also bridge the semantic gap between multiscale features. Next, we use the transformer decoder to map the learned high-level semantic information into the pixel space to refine the original features. In addition, we use the context extraction module (CEM) to obtain the local and global associations of feature maps. Finally, the addition of an attention aggregation module (AAM) can effectively combine the feature information at different scales. Extensive experiments on three public change detection datasets show that the proposed method has advantages over other methods in terms of both visual interpretation and quantitative analysis.**

***Index Terms*— Attention mechanism, deep learning, remote sensing image change detection, transformers.**

I. Introduction

REMOTE sensing image change detection refers to the process of change region extraction for multiple remote sensing images acquired at different times and from the same area. Currently, change detection is widely used in the fields of land planning [1], urban expansion [2], disaster assessment [3], and ecological monitoring [4]. Therefore, it is of great practical significance to develop change information identification and extraction technology based on remote sensing images, in order to obtain timely information on surface cover changes. However, due to the complexity and difficulty of the remote sensing image change detection problem, even though a large number of research results have been proposed one after another, with certain experience in theory, methods, and applications, a universal method has not yet appeared.

Prior to using deep learning for change detection, traditional methods were commonly used to process remote-sensing images. Among the traditional methods, pixel-based methods [5], [6], [7], [8], [9], [10], [11] were mainly used in the early days. The thinking of these methods is to first obtain the difference image and then perform a pixel classification based on thresholding or clustering on the difference image to extract the change features. Since these methods process only the spectral information of the image and ignore the contexts, they are only applicable to low- and medium-resolution images. With the development of imaging technology, the spatial resolution of images is increasing and ground objects are more textured, but pixel-based methods cannot use this information effectively. As a result, object-based methods [12], [13], [14], [15], [16], [17] have been widely used. The core concept is to treat the object composed of multiple pixels of adjacent image elements as the minimum processing unit, which can make full use of the spectral and spatial features of the object. Although these methods can improve the change detection of high-resolution remote sensing images, they lack the consideration of the correlation between the object neighborhoods. Therefore, over-segmentation and under-segmentation problems easily occur, which affect the final change detection accuracy. Recently, researchers have proposed some scene-level change detection methods [18], [19], [20] and achieved better results. For example, Wang et al. [18] proposed a graph-based segmentation for a multivariate time series algorithm (MTS-GS), which avoids the interference of noise and seasonal changes by constructing a similarity matrix and a corresponding graph segmentation algorithm. Wu et al. [20] proposed a novel scene change detection method via kernel slow feature analysis (KSFA) and postclassification fusion, which combines independent scene and scene change detection to accurately determine change regions.

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In recent years, with the development of deep learning techniques, researchers have also introduced some classical deep neural network techniques (e.g., UNet [21] and ResNet [22]) to the change detection task to extract deeper representations. In the early days, the advent of fully convolutional networks [23] established a baseline model. Then, Siamese networks were used and became the architecture of choice for most change detection methods [24], [25], [26], [27], [28]. The spatial context information is extracted from each temporal remote sensing image to reveal the changes of interest. In addition, to refine the feature information and improve the change detection performance of the network, many new techniques have been introduced in recent networks, which include the application of attention mechanisms [29], [30], [31], [32], multiscale feature fusion [25], [33], and stacking more convolutional layers [34], [35].

Based on the previous analysis, although many deep learning models have been proposed and used for remote sensing image change detection, they still have some shortcomings. First, the existing methods ignore spatiotemporal context information in the feature encoding and decoding process and do not link high-level features with low-level features. This often leads to blurred target area contours and the appearance of small object misses in the predicted change maps. Second, since remote sensing images of different phases are imaged by seasons or lighting conditions, the feature targets with the same semantic concept at different times and different spatial locations exhibit different spectral characteristics. These features introduce a large amount of noise interference to the remote sensing images, which makes the change detection results of most methods produce false detection, i.e., there are pseudochanges.

To address the above-mentioned problems, in this article, we propose an attention-guided multiscale context aggregation network (AMCA) for remote sensing image change detection. First, we use a Siamese backbone network to obtain the features at different levels of the two remote sensing images and further enhance the feature maps by a fully attentional pyramid module (FAPM). Then, we employ the dense feature fusion module (DFFM) to fuse the bitemporal features connected together at different levels to exclude the interference of irrelevant factors. Next, we use channelwise cross fusion transformer (CCT) [36] to capture the semantic features under different channels. And the semantic features at different scales are fused more effectively by channelwise cross attention (CCA) [36] to address the semantic gaps that exist between features at different levels. Subsequently, we pass the fused feature information through the transformer decoder (TD) to further enhance the spatiotemporal context information and generate a refined feature map. In addition, we use the context extraction module (CEM) to obtain the local and global correlation of the feature maps. And the original feature information and the semantic information are integrated by the attention aggregation module (AAM). Finally, we feed the computed feature difference maps into the prediction module (PM) to generate the predicted change results.

The main contributions of this article can be summarized as follows.

1) In this research, we design a novel remote sensing image change detection network by effectively combining convolutional neural networks (CNN) and transformer using knowledge from several specific domains. And extensive experiments have been conducted on three open change detection datasets to demonstrate the better performance of the proposed method.
2) We use the FAPM to determine the similarity between pixels in the feature map and use full attention to enhance the extraction and fusion of multiscale features. Meanwhile, in order to strengthen the spatiotemporal connection of the same type of objects, we introduce DFFM to fuse the bitemporal features and get the feature information of the change region.
3) We introduce CCT, which can effectively fuse channel features of different semantic patterns in multiple scales to capture more complex channel-based dependencies. The addition of CCA can address the semantic gap that exists between low-level and high-level features. In addition, to map the acquired high-level semantic features to pixel space, we use a modified TD for optimization to further enhance the change detection.
4) In order to capture local and global features, we use CEM to obtain a more accurate feature representation. Moreover, we use AAM to fuse the detailed information of low-level features and the semantic information of deep-level features and perform asymmetric filtering to extract more accurate and finer features.

The rest of this article is organized as follows. Section II is related work, describing the development status of remote sensing image change detection methods and the models based on transformer and attention applied in change detection. Section III describes the proposed method in detail. Section IV describes the experimental results. Section V draws the conclusion.

II. Related Work

A. Deep Learning-Based Change Detection

Due to the excellent performance of deep learning techniques in classification [37], [38], [39], [40], [41], segmentation, and other vision tasks [42], [43], many scholars have applied them to the field of change detection [34], [44]. The basic process of deep learning-based change detection methods is to extract features from the input image pairs and fuse the extracted features together. The change detection task is then converted to a pixel-by-pixel classification task to generate the change map.

For remote sensing change detection, most of the existing methods use a U-shaped encoder–decoder structure for feature extraction, where the encoder consists of an early fusion [32], [45], [46] or late fusion [24], [25], [27], [34], [47], [48] framework. Early fusion refers to the concatenated bitemporal image as the input, while late fusion refers to the feature extraction of the two images separately. Moreover, according on whether the weights are shared or not, late fusion can be divided into Siamese and pseudo-Siamese structures. For example, Daudt et al. [24] used Siamese structures for feature extraction and introduced three fully CNN for change

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detection. Their main operation is to connect the image in advance to the encoder or to pass the connection/difference features to the encoder. Fang et al. [27] proposed a new change detection structure by combining the Siamese network and nested U-Net [49]. The network performs feature extraction on bitemporal images by means of Siamese encoders and uses the channel attention module to refine the features. Liu et al. [47] constructed a dual-task constrained deep Siamese convolutional network (DTCDSCN) consisting of two semantic segmentations and a change detection network. They took the SE-ResNet network [50] as an encoder and took building extraction as an auxiliary task to additionally learn distinguishable object-level features. Zhang et al. [25] proposed a deeply supervised image fusion network (DSIFN) based on a pseudo-Siamese structure for extracting deep features and difference recognition. However, most networks are mainly aimed at obtaining finer features and lack modeling of foreground change objects, resulting in blurred change regions and more false detections.

Unlike existing methods that directly process the spatial pixel information in features, we use multiscale feature fusion and attention mapping to model the global semantic information. Specifically, we first fuse the bitemporal features efficiently and aggregate the fused features into several compact semantic tokens. Then, we integrate the global context into these markers so that they contain high-level semantic information while enhancing the original pixel features. In addition, we further strengthen the connection between different levels of features, which makes the change detection results more accurate.

B. Applications of Transformer Model

The transformer is a novel deep-learning method proposed by Google in 2017. It discards the traditional CNN and recurrent neural network (RNN), and the whole network structure consists entirely of attention mechanisms. Due to its excellent performance and wide applicability, researchers have gradually introduced transformer to computer vision tasks, including image classification [51], [52], target detection [53], [54], image segmentation [55], [56], and image enhancement [57], [58], [59].

Given the power of transformer to model context information and the absence of human-defined perceptual bias, researchers have applied it to the field of remote sensing image change detection [60], [61], [62], [63], [64], [65]. For example, Chen et al. [60] proposed a bitemporal image transformer, which uses semantic tokens to represent the features of the input image and models the global information in the image based on these tokens. Li et al. [61] combined UNet and transformer to connect multiscale features via jump connections to learn local-global semantic features. Bandara and Patel [62] combined a hierarchically structured transformer encoder with a multilayer-aware [multilayer perceptron (MLP)] decoder in a Siamese network to efficiently present multiscale remote details for accurate change detection. Feng et al. [63] proposed an intrascale cross interaction and interscale feature fusion network that explicitly taps into the fusion capabilities of CNN and transformer. Zhang et al. [64] designed a pure transformer network (SwinSUNet) with Siamese U-Net [21] structure to solve the change detection problem. The network contains encoders, fusers, and decoders, all of which use the Swin transformer block [66] as the base unit. In this article, we also apply the transformer to our change detection task, in order to obtain the global semantic information of the input image in the network and capture the dependencies between pixels.

C. Attentional Mechanism

The nature of the attentional mechanism is a dispositional resource allocation mechanism. It can effectively capture global relationships and focus limited energy on important locations from a global perspective, resulting in more useful information.

So far, researchers have proposed various attention modules in the field of computer vision. Hu et al. [50] used SENet to automatically obtain the importance of each feature channel. The convolutional block attention module (CBAM) [67] combines two dimensions, feature channel and feature space, to further develop SENet. Wang et al. [68] applied the idea of CBAM to change detection. They combined CBAM with a multiscale transformer to improve the detection quality of different remote sensing images. Subsequently, Wang et al. [69] proposed an efficient channel attention (ECA) module. This module is a lightweight module that can obtain better feature information while reducing the number of model parameters. The attention module proposed above is mainly biased toward the distribution of weights. Later, the application of the self-attention mechanism in the nonlocal module [70] allows not only to focus on the weight distribution but also to strengthen the context connection. Chen and Shi [28] used the self-attention module to calculate the attention weights of any two pixels between different times and positions to obtain better detection results. Fu et al. [71] proposed a dual-attention module (DAM) to obtain more discriminative image features. Chen et al. [29] applied this idea to remotely sensed images. They used a DAM to capture long-range dependencies to obtain a more discriminative feature representation.

In this article, we introduce a new attentional module. It captures the complete attentional similarity between any two channels and the associated spatial locations by means of a self-attention mechanism. At the same time, it can effectively suppress redundant information and improve the self-adaptive ability of large-scale changes.

III. Methodology

In this section, we will introduce the specific structure of the proposed change detection network. First, we introduce the overall framework and operation process of the network. Then, the functions of the modules that make up the network model are elaborated. Finally, the loss function we use is presented.

A. Network Architecture

The overall structure of the proposed change detection network is shown in Fig. 1. First, we use a Siamese feature extraction backbone to extract the multilevel features of each

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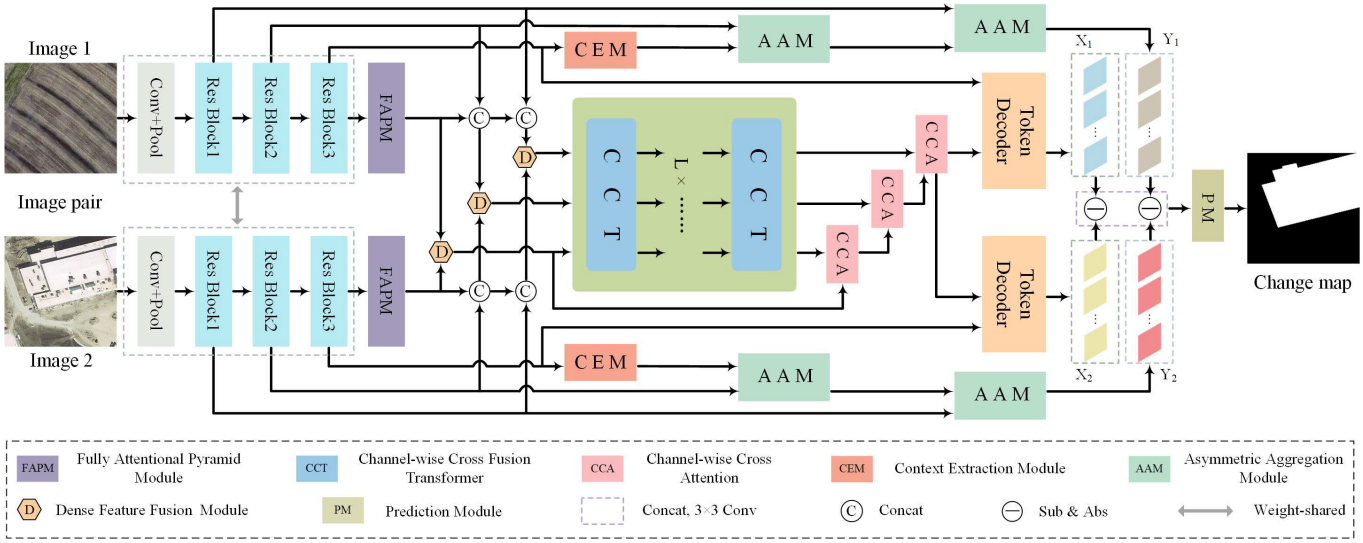


Fig. 1. Overall structure of the proposed network model. The input remote sensing images are processed level by level using the Siamese backbone network and the extracted deep semantic features are enhanced by the FAPM. DFFM is used to better fuse the bitemporal feature information. Feature information at different scales is adaptively processed by CCT and CCA, and the acquired semantic features are refined by using TD. CEM is used to fully extract multiscale information from the feature maps, and AAM is used to integrate low-level features with deep-level features.

input image. Among them, the backbone network is composed of two modified and weight-sharing ResNet18 [22]. Then, the features extracted from the two networks are fed into the FAPM separately, thus further enhancing the extraction and fusion of multiscale features. Immediately after, we employ DFFM to fuse features at different scales in bitemporal remote sensing images to generate changing features. Subsequently, the fused features at different scales are fed into CCT [36], so that the global semantic information in different channels can be obtained. And, to better fuse the semantic features between different scales of CCT output, we use CCA [36]. Meanwhile, we divide the generated feature mapping containing rich semantic information into two parts and feed them to the improved TD to obtain the refined pixel-level features. In addition, CEM computes local and global correlations of feature mappings to obtain a more accurate feature representation. Then, the feature information between the low-level and deep-level is fused using AAM to obtain the refined features. Next, the bitemporal features extracted by AAM and TD are pixel-subtracted to obtain different feature difference maps, respectively. Finally, the two feature maps are concatenated together and then undergo convolution operation and PM to generate the final change map.

We show the algorithmic flow of the proposed method in algorithm 1.

B. Fully Attentional Pyramid Module

Since ground objects have different feature information at different scales, effective extraction and fusion of features at multiple scales and large perceptual fields are necessary. Although researchers using some feature pyramid network (FPN) [72]-based methods can extract features effectively, they cannot handle long-distance dependencies in the feature map and are more difficult to extract for large object features. Here, we propose a FAPM based on the FPN. Fig. 2 shows the overall composition structure of the FAPM. From Fig. 2, it can

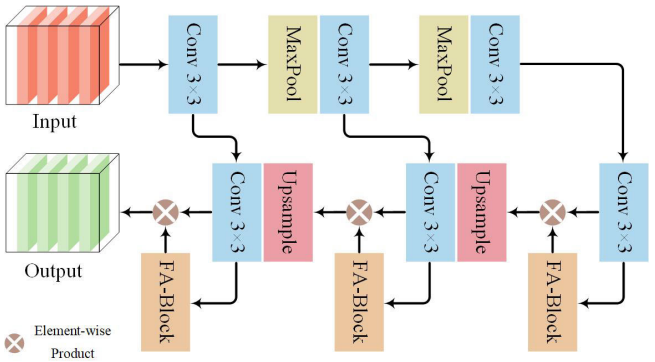


Fig. 2. Detailed structure of the FAPM.

be seen that FAPM mainly consists of convolutional layers and fully attentional blocks (FABs) [73]. And, FABs are added to the upsampling stage of FAPM. Our FAPM can enhance the feature map by judging the similarity between pixels and extracting and fusing features using a fully attentional approach. And it considers the spatial information between any two positions in the feature map, so it can further strengthen the intraclass correlation and improve the performance of interclass separation. In addition, the FAB we use is a new type of nonlocal block. It utilizes the global average set result as a global context prior, introducing spatial interactions into the channel nonlocal mechanism. This block effectively solves the attention deficit problem in the traditional nonlocal block and improves the feature discrimination ability.

Fig. 3 shows the detailed structure of the FAB. It can be seen that the input feature map $\boldsymbol{F}_{\text{in}} \in R^{C\times H\times W}$ (C is the number of channels, H and W are the spatial dimensions of the input tensor) is fed into the three branches. First, for the bottom branch, $\boldsymbol{F}_{\text{in}}$ is passed through a construction layer to obtain the global context feature map $\boldsymbol{Q} \in R^{(H+W)\times C\times S}$. Immediately after, the same operation steps are performed for the first two branches to generate feature map $\boldsymbol{K} \in R^{(H+W)\times C\times S}$ and

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Algorithm 1 AMCA

Input: Bitemporal remote sensing images $\boldsymbol{I} = \{(\boldsymbol{I}_1, \boldsymbol{I}_2)\}$
Output: Predicted change map $\boldsymbol{M}_{\text{new}}$

// Use feature extraction backbone to obtain different levels of features
for i in $\{1, 2\}$ **do**
  for j in $\{1, 2, 3\}$ **do**
    $\boldsymbol{F}_{ij} = \text{FE_Backbone}(\boldsymbol{I}_i)$
  end for
end for
// Use FAPM to process deep features and connect features at different levels
for i in $\{1, 2\}$ **do**
  $\boldsymbol{C}_i = \text{FAPM}(\boldsymbol{F}_{i3})$
  $\boldsymbol{B}_i = \text{Concat}(\boldsymbol{C}_i, \boldsymbol{F}_{i2})$
  $\boldsymbol{A}_i = \text{Concat}(\boldsymbol{B}_i, \boldsymbol{F}_{i1})$
end for
// The different levels of bitemporal features are fused using multiple DFFMs
$\boldsymbol{C} = \text{DFFM}(\boldsymbol{C}_1, \boldsymbol{C}_2)$
$\boldsymbol{B} = \text{DFFM}(\boldsymbol{B}_1, \boldsymbol{B}_2)$
$\boldsymbol{A} = \text{DFFM}(\boldsymbol{A}_1, \boldsymbol{A}_2)$
// Use CCT and CCA to process multilevel features to obtain rich semantic information
$\boldsymbol{A}_{\text{new}}, \boldsymbol{B}_{\text{new}}, \boldsymbol{C}_{\text{new}} = \text{CCT}(\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C})$
$\boldsymbol{T} = \text{CCA}(\boldsymbol{A}_{\text{new}}, \boldsymbol{B}_{\text{new}}, \boldsymbol{C}_{\text{new}})$
$\boldsymbol{T}_1, \boldsymbol{T}_2 = \text{Split}(\boldsymbol{T})$
// Use improved transformer decoder to generate refined pixel-level features
for i in $\{1, 2\}$ **do**
  $\boldsymbol{X}_i = \text{TD}(\boldsymbol{F}_{i3}, \boldsymbol{T}_i)$
end for
// Use CEM and AAM to process local and global information of feature maps
for i in $\{1, 2\}$ **do**
  $\boldsymbol{E}_i = \text{CEM}(\boldsymbol{F}_{i3})$
  $\boldsymbol{O}_i = \text{AAM}(\boldsymbol{E}_i, \boldsymbol{F}_{i2})$
  $\boldsymbol{Y}_i = \text{AAM}(\boldsymbol{O}_i, \boldsymbol{F}_{i1})$
end for
// Obtain the feature difference map
$\boldsymbol{X} = |\boldsymbol{X}_1 - \boldsymbol{X}_2|$
$\boldsymbol{Y} = |\boldsymbol{Y}_1 - \boldsymbol{Y}_2|$
$\boldsymbol{M} = \text{Concat}(\boldsymbol{X}, \boldsymbol{Y})$
// Get the final change map by PM
$\boldsymbol{M}_{\text{new}} = \text{PM}(\boldsymbol{M})$

feature map $\boldsymbol{V} \in R^{(H+W)\times C\times S}$, respectively. Then, we perform the affinity operation on $\boldsymbol{Q}$ and $\boldsymbol{K}$ such that each spatial location in the feature map receives feature responses from the global in the same row and column, i.e., the full attentions $\boldsymbol{A} \in R^{(H+W)\times C\times C}$ are obtained. The affinity operation is defined as follows:

$$\boldsymbol{A}_{i,j} = \frac{\exp\left(\boldsymbol{Q}_i \cdot \boldsymbol{K}_j\right)}{\sum_{i=1}^{C} \exp\left(\boldsymbol{Q}_i \cdot \boldsymbol{K}_j\right)} \tag{1}$$

where $\boldsymbol{A}_{i,j} \in \boldsymbol{A}$ denotes the degree of correlation between the ith and jth channels at a particular spatial location. Next,

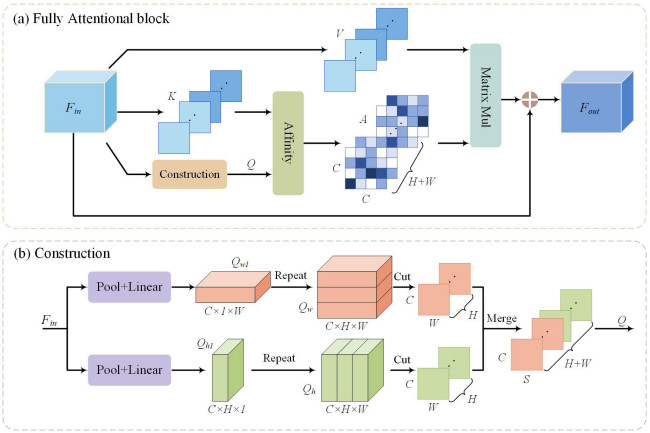


Fig. 3. Illustration of FAB. The ⊕ represents elementwise addition.

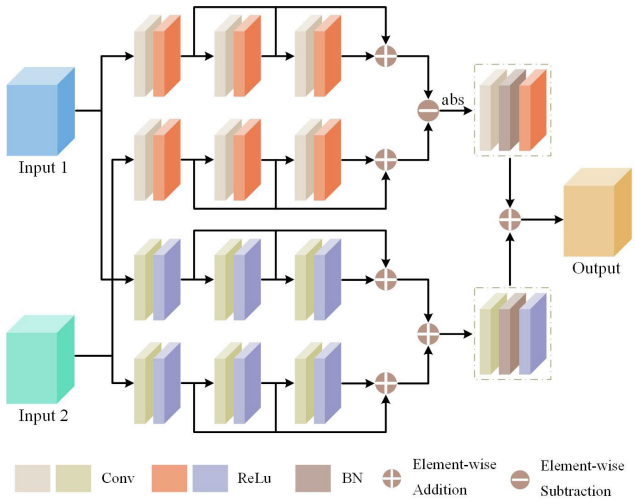


Fig. 4. Specific structure of DFFM.

we perform a matrix multiplication operation on $\boldsymbol{A}$ and $\boldsymbol{V}$, i.e., we update each channel information in the feature map with the generated full attention. Finally, we perform the elementwise sum operation of the result of the matrix operation with $\boldsymbol{F}_{\text{in}}$ to obtain the output feature $\boldsymbol{F}_{\text{out}} \in R^{C\times H\times W}$. The running process is shown in the following:

$$\boldsymbol{F}_{\text{out}}^{j} = \gamma \sum_{i=1}^{C} \boldsymbol{A}_{i,j} \cdot \boldsymbol{V}_j + \boldsymbol{F}_{\text{in}}^{j} \tag{2}$$

where $\boldsymbol{F}_{\text{out}}^{j}$ denotes the feature vector in the output feature map $\boldsymbol{F}_{\text{out}}$ of the jth channel map, and γ is a scale parameter.

C. Dense Feature Fusion Module

Due to the complex and diverse background information of bitemporal remote sensing images, and the bias in spatial location and color. There are still many misalignments between the bitemporal features we extracted through the Siamese network, which are not conducive to detecting changing objects. In order to better fuse the bitemporal features and obtain a clear region of change, we introduce DFFM into the network. Fig. 4 shows the detailed structure of the DFFM. It can be seen that the DFFM consists of a sum branch at the bottom

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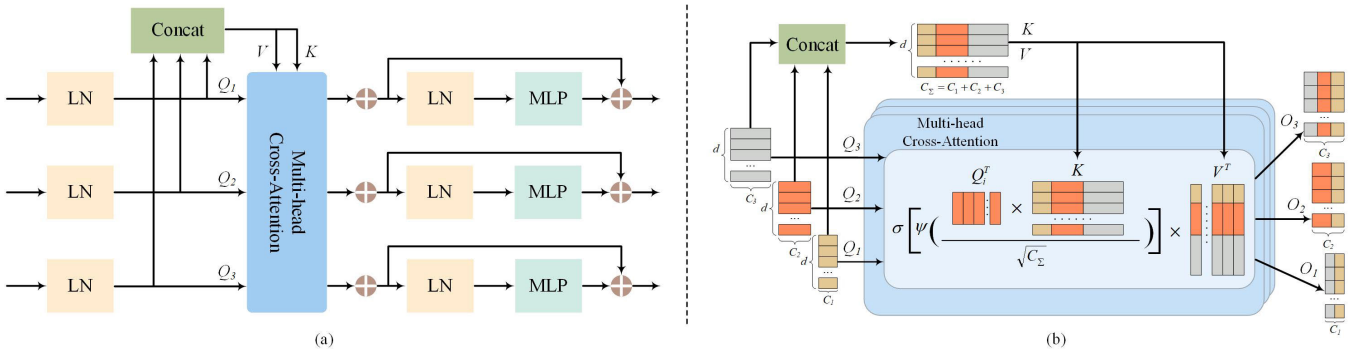


Fig. 5. Illustration of CCT. (a) Overall structure. (b) Detailed flow of the multihead channel cross-attention block.

and a difference branch at the top. The sum branch is used to capture more details and edge information in the feature map, and the difference branch is used to generate regions of change. Each branch consists of two densely connected data streams with shared weights. Moreover, each data stream contains three convolutional layers of 3×3 and three ReLU activation functions. In addition, the convolutional layers and ReLU in both sum and difference branches are the same. Formally, the sum branch and the difference branch are defined as

$$\boldsymbol{F}_1 = H_l\left(\left[\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_{l-1}\right]\right) \tag{3}$$

$$\boldsymbol{F}_2 = H_l\left(\left[\boldsymbol{y}_1, \boldsymbol{y}_2, \ldots, \boldsymbol{y}_{l-1}\right]\right) \tag{4}$$

$$\boldsymbol{F}_s = \boldsymbol{F}_1 + \boldsymbol{F}_2 \tag{5}$$

$$\boldsymbol{F}_d = |\boldsymbol{F}_1 - \boldsymbol{F}_2| \tag{6}$$

where $\boldsymbol{F}_s$ and $\boldsymbol{F}_d$ denote features generated by sum and difference branches, respectively, and $\boldsymbol{F}_1$ and $\boldsymbol{F}_2$ are the features generated from the bitemporal images after the dense concatenation operation. $[\cdot]$ represents the cascading operation, i.e., all the features $\boldsymbol{x}_i$ and $\boldsymbol{y}_i$ in front of the lth layer are connected separately. $H_l(\cdot)$ represents the nonlinear transformation function, which includes a series of ReLU and convolution operations. Finally, we add the results of the sum and difference branch outputs to obtain the output feature map $\boldsymbol{F}_o$. The running process can be expressed as follows:

$$\boldsymbol{F}_o = \delta(\text{conv}(\boldsymbol{F}_s)) + \delta(\text{conv}(\boldsymbol{F}_d)) \tag{7}$$

where $\delta(\cdot)$ is the ReLU function, and $\text{conv}(\cdot)$ denotes the 3×3 convolution layer.

Our DFFM can aggregate multiple features in each data stream to avoid pseudovariation caused by feature misalignment. In addition, since there are multiple residual connection operations in each stream, the features generated later can correct the previous one to some extent, thus making the generated feature map have a more accurate change region.

D. Channel Transformer

To further capture nonlocal semantic dependencies in multiscale features and improve the modeling ability of the network for global dependencies, we introduce the channel transformer [36]. The model consists of two main components, CCT and CCA.

1) Channelwise Cross Fusion Transformer: CCT fuses multiscale context with cross-attention from a channel perspective to achieve local cross-channel interaction. And CCT can adaptively fuse channel features focusing on different semantic patterns together, thus improving the segmentation performance of the network model. Fig. 5(a) shows the overall composition of the CCT. First, we tokenize the features $\boldsymbol{D}_i (i = 1, 2, 3)$ output from the three DFFMs, i.e., we reshape them into 2-D patch sequences with the size of p, $p/2$, and $p/4$, respectively. In this process, we make sure that the dimensions of the channels remain the same. Then, we join the tokenized features $\boldsymbol{T}_i (i = 1, 2, 3)$ together to obtain $\boldsymbol{T}_\Sigma$. Formally

$$\boldsymbol{T}_\Sigma = \text{Concat}(\boldsymbol{T}_1, \boldsymbol{T}_2, \boldsymbol{T}_3). \tag{8}$$

Subsequently, we feed these tokens $\boldsymbol{T}_i (i = 1, 2, 3)$ as query $(\boldsymbol{Q}_i)$ into the multihead channel cross-attention module. Finally, the channels and dependencies are processed by a multilayer perceptron with a residual structure for further feature refinement.

Fig. 5(b) shows the detailed flow of the multihead channel cross-attention block. As shown in Fig. 5(b), the block takes three tokens $\boldsymbol{T}_i$ as queries $(\boldsymbol{Q}_i)$ and the joined tokens $\boldsymbol{T}_\Sigma$ as key $(\boldsymbol{K})$ and value $(\boldsymbol{V})$. It can be expressed as

$$\boldsymbol{Q}_i = \boldsymbol{T}_i W_{\boldsymbol{Q}_i}, \quad \boldsymbol{K} = \boldsymbol{T}_\Sigma W_{\boldsymbol{K}}, \quad \boldsymbol{V} = \boldsymbol{T}_\Sigma W_{\boldsymbol{V}} \tag{9}$$

where $W_{\boldsymbol{Q}_i} \in R^{C_i \times d} (i = 1, 2, 3)$, $W_{\boldsymbol{K}} \in R^{C_\Sigma \times d}$, and $W_{\boldsymbol{V}} \in R^{C_\Sigma \times d}$ denote the weights corresponding to query, key, and value, respectively, d is the number of patches (length of the sequence), and $C_i (i = 1, 2, 3)$ denotes the channel sizes of the different input layers. In the network model, we set as: $C_1 = 64$, $C_2 = 128$, $C_3 = 256$. Then, we generate the similarity matrix $\boldsymbol{M}_i$ by $\boldsymbol{Q}_i \in R^{C_i \times d}$, $\boldsymbol{K} \in R^{C_\Sigma \times d}$ and $\boldsymbol{V} \in R^{C_\Sigma \times d}$, and use $\boldsymbol{M}_i$ to weight the value $(\boldsymbol{V})$ by the cross-attention (CA) mechanism. The formula is expressed as follows:

$$\begin{aligned} \text{CA}_i &= \boldsymbol{M}_i \boldsymbol{V}^T \\ &= \sigma\left[\psi\left(\frac{\boldsymbol{Q}_i^T \boldsymbol{K}}{\sqrt{C_\Sigma}}\right)\right]\boldsymbol{V}^T \\ &= \sigma\left[\psi\left(\frac{W_{\boldsymbol{Q}_i}^T \boldsymbol{T}_i^T \boldsymbol{T}_\Sigma W_{\boldsymbol{K}}}{\sqrt{C_\Sigma}}\right)\right] W_{\boldsymbol{V}}^T \boldsymbol{T}_\Sigma^T \end{aligned} \tag{10}$$

where $\psi(\cdot)$ denotes instance normalization, $\sigma(\cdot)$ is the softmax function, and C_Σ denotes the number of channels after

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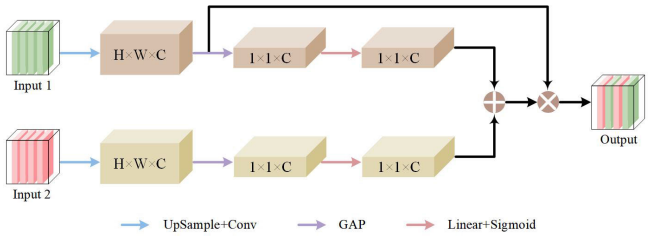


Fig. 6. Description of the CCA module. ⊕ represents elementwise addition. ⊗ represents elementwise product.

connection. Unlike the original self-attention along the patch-axis, we perform attention operations along the channel-axis. And we normalize each similarity matrix to ensure that the gradient can be propagated smoothly. Immediately after, the output of the multihead channel cross attention is obtained as follows:

$$\mathrm{MCA}_i = \frac{\mathrm{CA}_i^1 + \mathrm{CA}_i^2 +, \ldots, +\mathrm{CA}_i^N}{N} \tag{11}$$

where N is the number of attention heads. After that, the channel relations are processed using a multilayer perceptron to obtain the output results. The formula is expressed as follows:

$$\boldsymbol{O}_i = \mathrm{MCA}_i + \varsigma\left(\boldsymbol{Q}_i + \mathrm{MCA}_i\right) \tag{12}$$

where $\varsigma(\cdot)$ denotes the multilayer perceptron. Finally, we repeat the operation in (5) three times to build a three-layer transformer, which generates three different scales of outputs $\boldsymbol{O}_1$, $\boldsymbol{O}_2$, and $\boldsymbol{O}_3$.

2) Channelwise Cross Attention: CCA can guide the channel information and effectively filter the connected feature information to remove the ambiguity between different scale features. Fig. 6 shows the detailed structure of the CCA. First, we perform upsampling and convolution operations on two different input features, respectively. Then, the features after these two upsampling are compressed by the global average pooling (GAP) layer to embed the global spatial information. Immediately after, considering the effect of dimensionality reduction on learning channel attention, we use a single linear layer and Sigmoid activation function to construct the channel attention map. And the two attention maps are connected to generate an attention mask that can represent the correlation between these two feature channels. Finally, we multiply the connected vectors with the input features to obtain the output.

E. Transformer Decoder

We tokenize the features' output by the channel transformer to obtain the set $\boldsymbol{S}_i (i = 1, 2)$ of tokens with context information, in which the target change information is also contained. Now, we need to project this high-level semantic information into pixel space to obtain pixel-level features of the image. To achieve this, we use a modified Siamese TD to extract image features, the structure of which is shown in Fig. 7. The module has N_D layers, each containing a multicross attention (MCA) block and a multilayer perceptron (MLP) block. First, the feature map $\boldsymbol{F}_i (i = 1, 2)$ is represented by a sequence. Then, the improved TD obtains a new feature map $\boldsymbol{X}_i (i = 1, 2)$ based on the association of each pixel with each semantic token $\boldsymbol{S}_i (i = 1, 2)$. Here, we consider $\boldsymbol{F}_i$ as a query vector $\boldsymbol{Q}$, and the newly obtained semantic vector $\boldsymbol{S}_i$ as a key vector $\boldsymbol{K}$ and a value vector $\boldsymbol{V}$. Thus, the process is expressed as

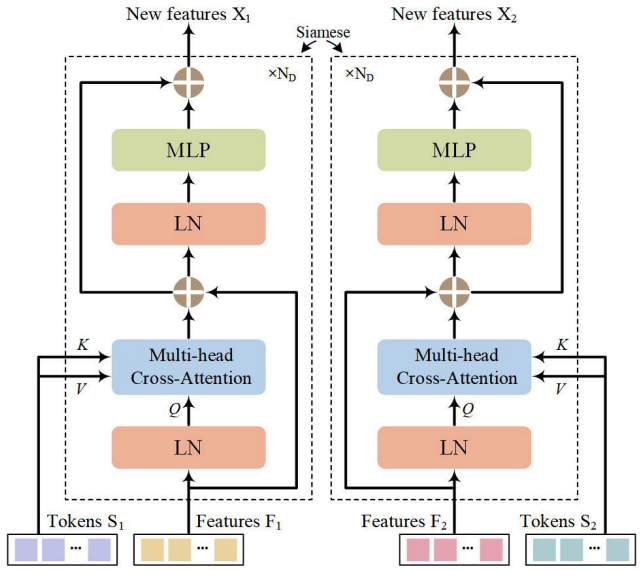


Fig. 7. Illustration of the TD module. ⊕ represents elementwise addition.

$$\mathrm{MCA}(\boldsymbol{F}_i, \boldsymbol{S}_i) = \mathrm{Concat}(\mathrm{head}_1, \ldots, \mathrm{head}_n)\boldsymbol{W}^O \tag{13}$$

where $\mathrm{head}_i (i = 1, 2, \ldots, n)$ denotes the output of each attention head, $\boldsymbol{W}^O$ is the projection matrix of the last linear layer in the MCA.

F. Context Extraction Module

Considering the complex and diverse contents in the input bitemporal remote sensing images and the inconsistent size of buildings in the images. We use CEM in the network to further extract multiscale features that are both global and local and to improve the feature representation capability of the network. Fig. 8 shows the overall composition architecture of the CEM. First, we use a convolutional layer of 1×1 to perform a dimensionality reduction operation on the input feature map $\boldsymbol{F}_i (i = 1, 2)$ to obtain a new feature map $\boldsymbol{F}'_i (i = 1, 2)$. Then, $\boldsymbol{F}'_i$ is fed in parallel to several attention-based context blocks (ACBs) of different scales to obtain features $\boldsymbol{M}_i^{Sj} (i = 1, 2; j = 1, 2, 3)$ of different scales, where S denotes the scale vector. Next, multiple scale features $\boldsymbol{M}_i^{Sj}$ are aggregated together and connected with $\boldsymbol{F}'_i$. Formally

$$\boldsymbol{M}_i = \mathrm{Concat}\left(\boldsymbol{M}_i^{S1}, \boldsymbol{M}_i^{S2}, \boldsymbol{M}_i^{S3}\right) \tag{14}$$

$$\boldsymbol{F}_i^* = \mathrm{Concat}\left(\boldsymbol{M}_i, \boldsymbol{F}'_i\right) \tag{15}$$

where $\boldsymbol{M}_i (i = 1, 2)$ denotes the features generated by the aggregation operation, and $\boldsymbol{F}_i^* (i = 1, 2)$ denotes the connected features. Finally, $\boldsymbol{F}_i^*$ is passed through a convolutional layer of 1×1 to integrate the channel information to obtain the output result.

The bottom of Fig. 8 shows the detailed structure of the ACB. As can be seen from Fig. 8, the ACB consists of two

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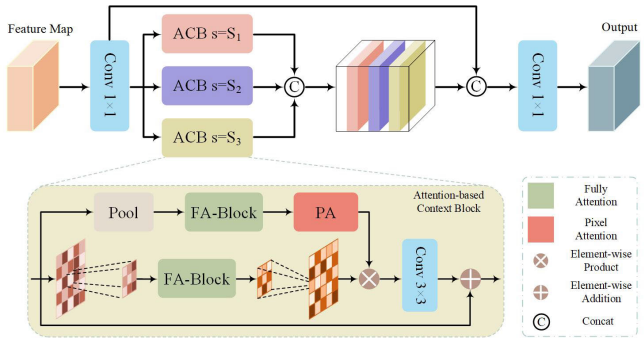


Fig. 8. Detailed structure of CEM.

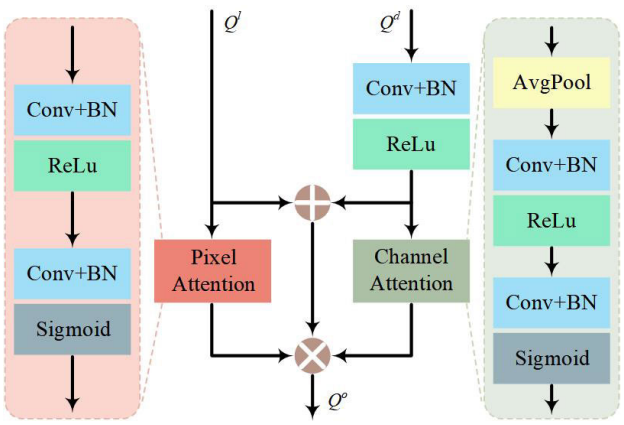


Fig. 9. Specific structure of the AAM. The ⊕ represents elementwise addition. The ⊗ represents an elementwise product.

branches. Among them, the upper branch is used to obtain global information between pixels, and the lower branch is used to calculate the local correlation of features. Specifically, in the upper branch, we first remove the redundant information from the feature map by using a maximum pooling layer. Then the FAB is used to analyze the context information in the pooled features in order to assist the network in discriminating the location of change targets. Among them, the FAB has been described in Section III-B. Immediately after, we pass the feature map through the pixel attention model, in order to integrate the feature information among all channels and get the feature map $\boldsymbol{G}_i(i=1,2)$ containing the global information. In the lower branch, we divide the feature map $\boldsymbol{F}_i'(i=1,2)$ into $s \times s$ patches, where the size of each patch is $w \times h$, and the weight is shared among all patches. Then, we use the FAB to calculate the dependency between pixels in each patch. In this way, the pixels of the same category can be concentrated and the discriminative results in the local range can be obtained. Subsequently, we pool the processed patches together to generate a new local association feature map $\boldsymbol{L}_i(i=1,2)$. Next, we upsample $\boldsymbol{G}_i(i=1,2)$ using bilinear interpolation and multiply the sampled features with $\boldsymbol{L}_i(i=1,2)$. Finally, the result of the multiplication is passed through a convolution layer of 3×3 and then summed with $\boldsymbol{F}_i'(i=1,2)$ to generate a new feature map $\boldsymbol{F}_i^{\text{new}}(i=1,2)$. The feature map has multiscale characteristics, and the map contains both global and detailed features, which reduces the loss of information and improves the change detection accuracy.

G. Attention Aggregation Module

In order to realize the change detection of small volume targets and obtain the edge information of complete change regions, we propose AAM. Specifically, AAM is the integration of low-level features and deep-level features from a fusion perspective. This module solves the problem of feature mismatch caused by separate constraints on low-level detail information and deep-level semantic information and improves the accuracy of the network for change detection of objects.

Fig. 9 shows the detailed structure of the AAM. First, we pass the low-level features $\boldsymbol{Q}_i^l$ through the pixel attention module to generate a pixel attention feature map $\boldsymbol{Q}_i^{\text{pa}}(i=1,2)$. It can be expressed as

$$\boldsymbol{Q}_i^{\text{pa}} = \sigma\left(\text{conv}\left(\delta\left(\text{conv}\left(\boldsymbol{Q}_i^l\right)\right)\right)\right) \tag{16}$$

where $\sigma(\cdot)$ is the Sigmoid function, $\delta(\cdot)$ is the ReLU function, and $\text{conv}(\cdot)$ denotes the 1×1 convolution layer. Then, we pass the deep feature $\boldsymbol{Q}_i^d(i=1,2)$ through a convolutional layer and a ReLU activation layer to generate the feature map $\boldsymbol{Q}_i^*(i=1,2)$. Next, $\boldsymbol{Q}_i^*$ is fed into the channel attention module to obtain the channel attention map $\boldsymbol{Q}_i^{\text{ca}}(i=1,2)$. It can be expressed as

$$\boldsymbol{Q}_i^* = \delta\left(\text{conv}\left(\boldsymbol{Q}_i^d\right)\right) \tag{17}$$

$$\boldsymbol{Q}_i^{\text{ca}} = \sigma\left(\text{conv}\left(\delta\left(\text{conv}\left(\text{AvgPool}\left(\boldsymbol{Q}_i^*\right)\right)\right)\right)\right) \tag{18}$$

where $\text{AvgPool}(\cdot)$ denotes the average pooling operation. Finally, we sum $\boldsymbol{Q}_i^*$ and $\boldsymbol{Q}_i^l$, and multiply the summation result with $\boldsymbol{Q}_i^{\text{pa}}$ and $\boldsymbol{Q}_i^{\text{ca}}$ to generate a new feature map $\boldsymbol{Q}_i^o(i=1,2)$. The formula is expressed as follows:

$$\boldsymbol{Q}_i^o = \left(\boldsymbol{Q}_i^* + \boldsymbol{Q}_i^l\right) \otimes \boldsymbol{Q}_i^{\text{pa}} \otimes \boldsymbol{Q}_i^{\text{ca}} \tag{19}$$

where $\boldsymbol{Q}_i^{\text{pa}}(i=1,2)$ denotes the feature map output by the pixel attention module, and $\boldsymbol{Q}_i^{\text{ca}}(i=1,2)$ denotes the feature map output by the channel attention module.

H. Network Details

We have obtained the high-level semantic features $\boldsymbol{X}_i(i=1,2)$ and $\boldsymbol{Y}_i(i=1,2)$ by TD module and AAM, so we only need to use simple FCN for change recognition. First, we perform the pixel absolute value subtraction operation on the feature maps $\boldsymbol{X}_i(i=1,2)$ and $\boldsymbol{Y}_i(i=1,2)$ to generate two feature difference maps: $\boldsymbol{X}, \boldsymbol{Y} \in R^{H_0 \times W_0 \times C}$ (H_0 and W_0 are the height and width of the original image, respectively). Then, $\boldsymbol{X}$ and $\boldsymbol{Y}$ are connected together and the results are convolved and upsampled to obtain the feature map $\boldsymbol{M} \in R^{H_0 \times W_0 \times C}$. Finally, $\boldsymbol{M}$ is passed through a PM to generate the predicted change probability map $\boldsymbol{M}_{\text{new}} \in R^{H_0 \times W_0 \times 2}$. The formula is expressed as follows:

$$\begin{aligned} \boldsymbol{M}_{\text{new}} &= \sigma(g(\boldsymbol{M})) \\ &= \sigma(g(\boldsymbol{X}+\boldsymbol{Y})) \\ &= \sigma(g(|\boldsymbol{X}_1 - \boldsymbol{X}_2| + |\boldsymbol{Y}_1 - \boldsymbol{Y}_2|)) \end{aligned} \tag{20}$$

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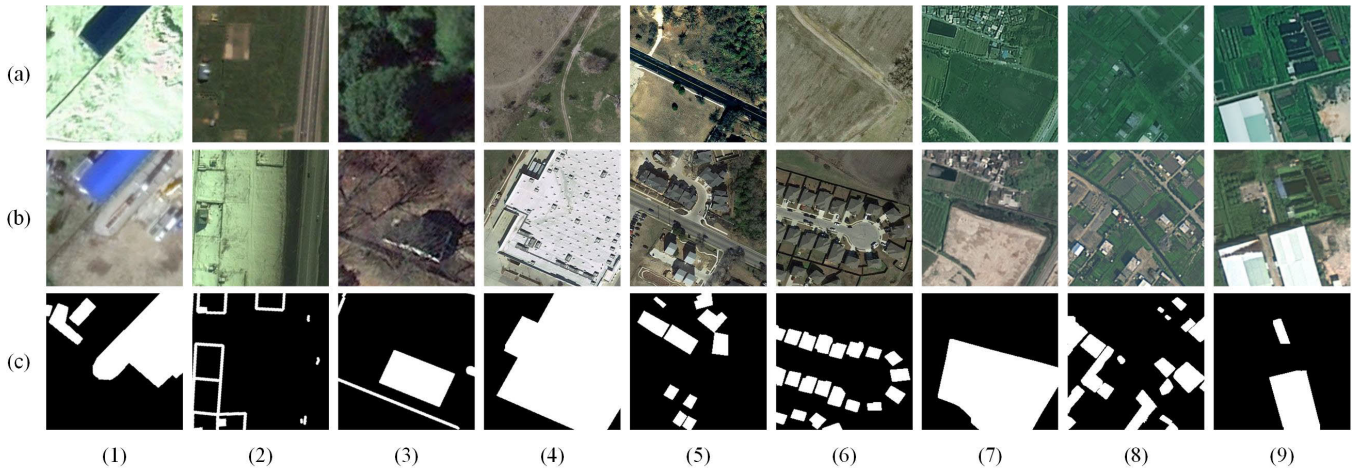


Fig. 10. Examples of the Season-Varying dataset (1)–(3), the LEVIR-CD dataset (4)–(6), and the CLCD dataset (7)–(9). (a) Remote sensing image at the moment of T1. (b) Remote sensing image at the moment of T2. (c) Ground Truth. White pixels indicate areas of change and black pixels indicate areas of no change.

where $g(\cdot): R^{H_0 \times W_0 \times C} \rightarrow R^{H_0 \times W_0 \times 2}$ denotes the change classifier, and $\sigma(\cdot)$ denotes the normalization operation of softmax function on the channel dimension output by the classifier. Our classifier uses two convolutional layers with a convolutional kernel of 3×3 and BatchNorm. And the number of channels output after two convolutions is 32 and 2, respectively.

For the selection of the loss function, in the training phase, we choose the minimized cross-entropy loss function to optimize the network parameters. This loss function is defined as

$$L = \frac{1}{H_0 \times W_0} \sum_{h=1,w=1}^{H,W} l(P_{hw}, Y_{hw}) \tag{21}$$

where $l(P_{hw}, y) = -\log(P_{hwy})$ denotes the cross-entropy loss, and Y_{hw} denotes the pixel coordinate (h, w) in the change reference map.

IV. Experiments

In this section, to verify the effectiveness of our proposed method, we choose three public change detection datasets for testing. First, we introduce the details of three datasets, namely, the Seasonal-Varying dataset [74], the LEVIR-CD dataset [28], and the CLCD dataset [75]. Second, we describe our implementation details, including experimental environment and parameter settings, evaluation metrics, and comparative methods. Then, the experimental results of our method and other change detection methods on three datasets are presented, and the experimental data are analyzed and summarized. Subsequently, we perform ablation experiments on the proposed method to prove its rationality and effectiveness. Finally, to further understand our method, we visualize the features of several important modules in the network model.

A. Datasets

1) Season-Varying Dataset: The Seasonal-Varying dataset [74] consists of 11 pairs of multisource remote sensing images, including seven pairs of seasonal variation images of size 4725 × 2700 pixels and four pairs of images of size 1900 × 1000 pixels. The resolution varies from 3 to 100 cm per pixel, with significant seasonal differences between bitemporal images. The original authors divided the dataset into the training dataset, validation dataset, and test dataset by cropping and rotating. The training dataset has 10 000 images, and the validation and test datasets both contain 3000 images, each with a size of 256 × 256. For comparison with other algorithms, we also use this data division method. Some examples in the Season-Varying dataset are shown in Fig. 10 (1)–(3).

2) LEVIR-CD Dataset: The LEVIR-CD dataset [28] consists of 637 pairs of high-resolution remote sensing images with a resolution of 0.5 m/pixel, each pair having a size of 1024 × 1024 pixels. The time span between image pairs is between five and 14 years, and there are a large number of significant land use changes, especially for new buildings. Due to the graphics processing unit (GPU) memory limitation, we use the same cropping strategy as Chen and Shi [28]. Each image in the dataset is cropped into 16 small blocks of 256 × 256 pixels in size with no overlapping regions. The training dataset consists of 7120 image pairs, the validation dataset consists of 1024 image pairs and the test dataset consists of 2048 image pairs. Some examples from the LEVIR-CD dataset are shown in Fig. 10 (4)–(6).

3) CLCD Dataset: The CLCD dataset [75] consists of 600 pairs of high-resolution remote sensing images of farmland changes with spatial resolution distributed between 0.5 and 2 m/pixel, and each pair is 512 × 512 pixels in size. The bitemporal images in this dataset were acquired in 2017 and 2019 by the Gaofen 2 in the Guangdong Province, China. We divide each image into nonoverlapping sub-images of 256 × 256 pixels in size and randomly divide them into the training dataset, validation dataset, and test dataset in the ratio of 6:2:2. Then, we rotate the training dataset by 90°, 180°, and 270° to achieve data enhancement. Finally, 5760 pairs of images were obtained for training, 480 pairs of images for validation, and 480 pairs of images for testing. Some examples from the CLCD dataset are shown in Fig. 10 (7)–(9).

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B. Implementation Details

1) Experimental Environment and Parameter Settings: Our experiments use Python 3.8 as the programming language and PyTorch 1.7.1 as the deep learning framework on which to build the network model. The models were trained, validated, and tested on a server with multiple NVIDIA GeForce RTX 2080Ti graphics cards and Ubuntu 18.04 LTS operating system.

Before training the model, we perform data enhancement processes on the input images, including cropping, rescaling, rotation, and Gaussian blurring. In the experimental process, we use stochastic gradient descent (SGD) with momentum as the optimization algorithm and set the parameters as momentum $= 0.9$ and weight decay $= 0.0005$. For all three training datasets, we set the learning rate to 0.01 and train 200 epochs for each dataset. Considering the limitation of video memory size, the batch size is set to 8 for all datasets.

2) Evaluation Metrics: In addition to visually comparing good and bad results, the change detection also quantitatively analyzes the algorithm's strengths and weaknesses by calculating the numerical indicators of the detection results and the reference map. Since the nature of the change detection task is a dichotomous problem, we adopt several commonly used evaluation metrics, which include precision (P), recall (R), F1-score (F1), overall accuracy (OA), and intersection over union (IoU). The calculation of these evaluation indicators is shown in the following:

$$P = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{22}$$

$$R = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{23}$$

$$\mathrm{F1} = \frac{2\mathrm{PR}}{P + R} \tag{24}$$

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}} \tag{25}$$

$$\mathrm{IoU} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP} + \mathrm{FN}} \tag{26}$$

where, true positive (TP) indicates the number of pixels where the change is correctly detected, false positive (FP) indicates the number of pixels that did not actually change but were incorrectly detected as having changed, true negative (TN) indicates the number of pixels for which no change is correctly detected, and false negative (FN) indicates the number of pixels that actually changed but were incorrectly detected as not having changed.

As can be seen from the above-mentioned formula, the higher the OA value indicates the higher the accuracy of the change detection results, the lower the P value indicates the more false detections, and the lower the R value indicates the more missed detections. F1 performs a weighted average reconciliation of precision and recall and is able to combine the two for evaluation. IoU indicates the degree of overlap between the predicted region and the labeled truth region, i.e., the ratio of the intersection and the concatenation between the predicted result and the ground truth pixels. Both F1 and IoU are comprehensive measurement metrics, and the larger their value, the better the change detection result.

3) Comparative Methods: To validate the effectiveness and superiority of the proposed network model, we compare our approach with ten recent deep learning-based change detection methods. These methods include FC-EF [24], FC-Siam-Conc [24], FC-Siam-Diff [24], DTCDSCN [47], DSIFN [25], SNUNet [27], region detail preserving network (RDPNet) [76], bitemporal image transformer (BIT) [60], ChangeFormer [62], and multiscale context aggregation network (MSCANet) [75]. The following is a brief description of each method.

FC-EF [24] is a fully convolutional network based on the U-Net structure. It connects bitemporal images together and extracts multiscale features of the images by ConvNet. FC-Siam-Conc [24] is an extension of FC-EF that splices shallow and deep features together in a jump connection structure. FC-Siam-Diff [24] is another extension of FC-EF. It calculates the difference between two shallow features in the jump join structure first and then splices the difference with the deep features. DTCDSCN [47] is a multiscale feature connectivity method, which uses the DAM to leverage the interdependencies between channels and spatial locations in order to extract more context features. DSIFN [25] fuses the extracted multilevel deep features with image difference features through an attention mechanism and uses deep supervision to train the intermediate layers of the network. SNUNet [27] is based on convolution, attention mechanisms, and multiscale networks. It uses the Siamese UNet++ network as a feature extraction tool and uses an integrated channel attention module to refine the features at different levels. RDPNet [76] is a RDPNet. It improves the attention of the network to boundaries and small regions through an effective training strategy and edge loss. BIT [60] is a transformer-based method that uses the transformer encoder–decoder structure to capture more context information in a compact token-based space-time. ChangeFormer [62] is a transformer-based Siamese network. It integrates a transformer encoder with a hierarchical structure and multilayer perception (MLP) decoder into a general change detection network framework to obtain long-range details. MSCANet [75] is a CNN-transformer method with multiscale context aggregation, which combines the advantages of CNN and transformer to achieve efficient and effective change detection.

C. Results and Analysis

We compare our method with the nine latest change detection methods mentioned earlier on the Season-Varying, LEVIR-CD, and CLCD datasets, respectively, and analyze the experimental results qualitatively and quantitatively. In addition, to ensure the accuracy of the results of each method and the fairness of the comparative experiments, we use the same dataset for training and testing for each method. The evaluation results for each method on each dataset are shown in the following.

1) Evaluation Results on the Season-Varying Dataset: The Seasonal-Varying dataset contains different kinds of change targets and various change scenarios. We select three typical scenes of buildings, roads, and land for visualization. Figs. 11, 12, and 13 show the change detection results of our

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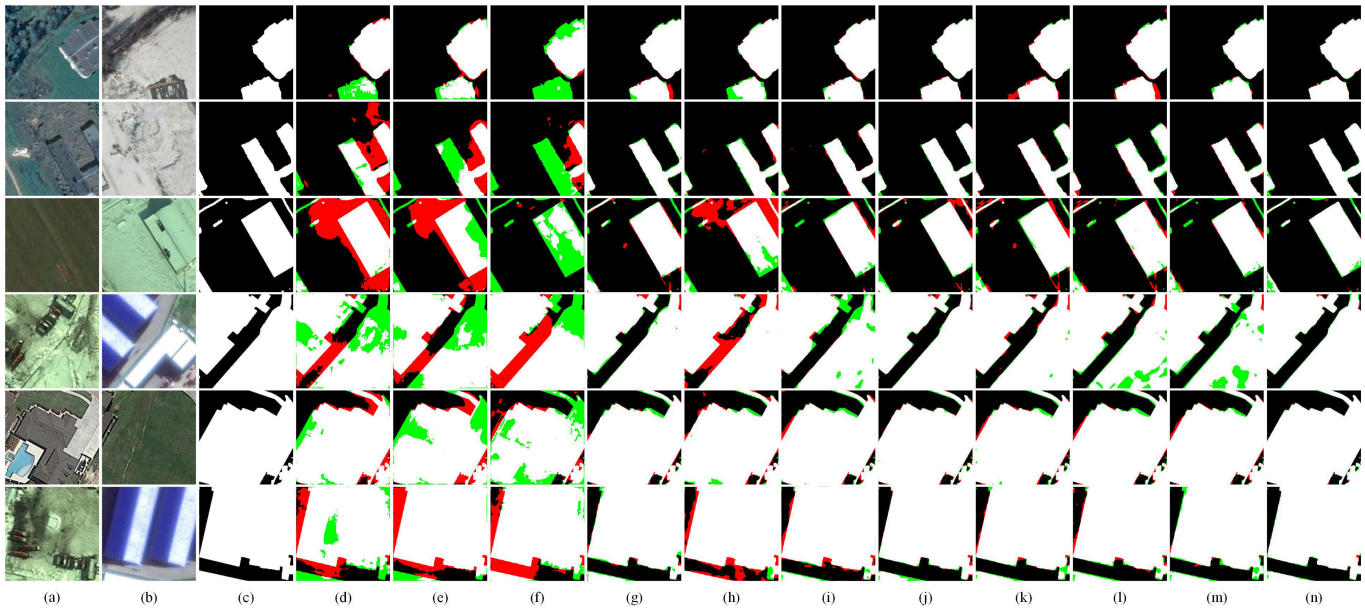


Fig. 11. Visualization results of different change detection methods in the Season-Varying dataset for building scenes. (a) and (b) Remote sensing images before and after the change. (c) Ground truth. (d) FC-Siam-Conc. (e) FC-Siam-Diff. (f) FC-EF. (g) SNUNet. (h) DTCDSCN. (i) DSIFN. (j) RDPNet. (k) MSCANet. (l) BIT. (m) ChangeFormer. (n) Ours. In the change map, white pixels represent TP, black pixels represent TN, red pixels represent FP, and green pixels represent FN.

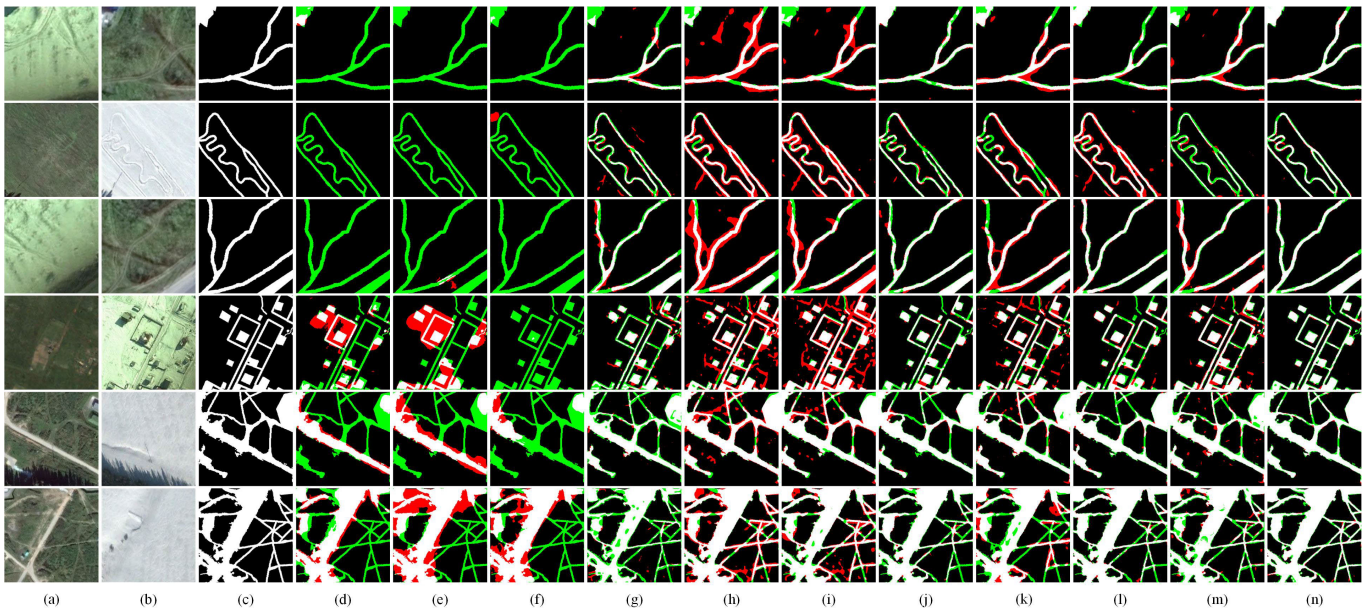


Fig. 12. Visualization results of different change detection methods in the Season-Varying dataset for road scenes. (a) and (b) Remote sensing images before and after the change. (c) Ground truth. (d) FC-Siam-Conc. (e) FC-Siam-Diff. (f) FC-EF. (g) SNUNet. (h) DTCDSCN. (i) DSIFN. (j) RDPNet. (k) MSCANet. (l) BIT. (m) ChangeFormer. (n) Ours. In the change map, white pixels represent TP, black pixels represent TN, red pixels represent FP, and green pixels represent FN.

method and other advanced methods in different scenarios, respectively.

Fig. 11 shows the results of change detection for different types of buildings. As can be seen, the first three rows of Fig. 11 show the changes in the small- and medium-sized buildings. The number of changed pixels is relatively small in the whole image, and most methods can roughly identify the changed area. The fourth and sixth rows of Fig. 11 show the changes in large buildings. The change areas are larger and more concentrated in the whole image, but the shape and outline of the change areas are different. The prediction results of FC-Siam-Conc, FC-Siam-Diff, FC-EF, and DTCDSCN have more false detections at the edge of the change area. The prediction results of DSIFN, BIT, and ChangeFormer lost some of the original image information, which led to the occurrence of missed detections. Compared with other methods, the predicted change maps of our method not only retain more detailed information in the small change areas but also segment the edges of the large change areas more clearly and more closely to the true labels.

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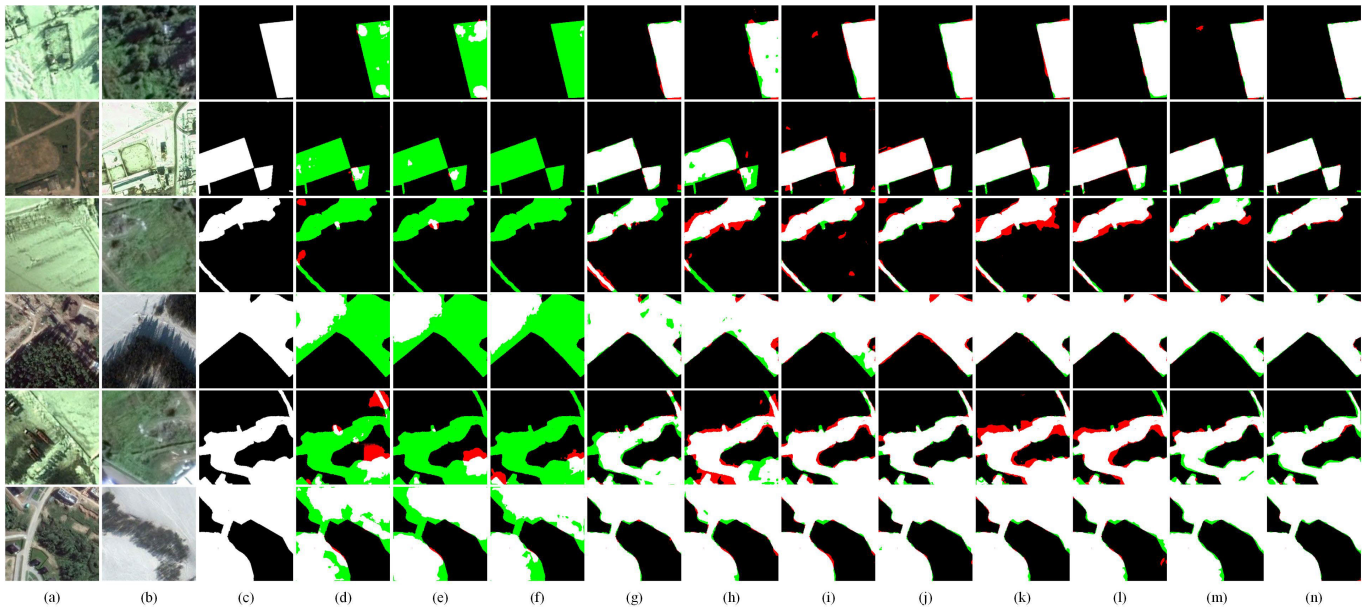


Fig. 13. Visualization results of different change detection methods in the Season-Varying dataset for land scenes. (a) and (b) Remote sensing images before and after the change. (c) Ground truth. (d) FC-Siam-Conc. (e) FC-Siam-Diff. (f) FC-EF. (g) SNUNet. (h) DTCDSCN. (i) DSIFN. (j) RDPNet. (k) MSCANet. (l) BIT. (m) ChangeFormer. (n) Ours. In the change map, white pixels represent TP, black pixels represent TN, red pixels represent FP, and green pixels represent FN.

Fig. 12 shows the change detection results for roads of different scales. As you can see, the labeled map contains both narrow roads with simple changes and rugged roads with complex changes. The prediction results of FC-Siam-Conc, FC-Siam-Diff, and FC-EF are less effective for road segmentation, with obvious missing regional targets in the change map. The prediction results of DTCDSCN, DSIFN, MSCANet, and ChangeFormer are more noisy and susceptible to pseudochanges, such as seasons and shadows, leading to erroneous judgments for roads that have not changed (e.g., rows 2 and 6 in Fig. 12). The prediction results of SNUNet, RDPNet, and BIT show regional discontinuities (e.g., rows 1 and 5 in Fig. 12). Compared with other methods, our method can better overcome the interference of pseudochanges and effectively deal with these irregular changes, and the regional connectivity has obvious advantages.

Fig. 13 shows the detection results of land change. As shown in Fig. 13, the predictions of FC-Siam-Conc, FC-Siam-Diff, and FC-EF are poor for both small and large land area changes, with very significant green areas and large missed areas in the change maps. There are obvious false detections and missed detections in the prediction results of DTCDSCN, DSIFN, MSCANet, and BIT, and the edge contour segmentation is not clear (e.g., rows 3 and 5 in Fig. 13). Although the remaining methods are all better at detecting the change fraction, our method detects the area of land change more accurately. For example, as can be seen in the second row in Fig. 13, our proposed model can discern more subtle changes in smaller change regions. Moreover, for large change regions (e.g., row 4 in Fig. 13), our network can recover more detailed information. Therefore, by qualitatively analyzing different scenarios in the Season-Varying dataset, it can be obtained that our method outperforms other advanced methods in terms of accuracy and predicted shape, locating the correct area of change, and having clear boundaries.

TABLE I

COMPARISON OF THE AVERAGE NUMERICAL METRICS OF DIFFERENT CHANGE DETECTION METHODS ON THE SEASON-VARYING DATASET. VALUES ARE DESCRIBED USING PERCENTAGES, AND THE BEST VALUES ARE HIGHLIGHTED IN BOLD

Method	P (%)	R (%)	F1 (%)	IoU (%)	OA (%)
FC-Siam-Conc	74.51	73.87	71.16	60.11	94.92
FC-Siam-Diff	78.25	65.76	69.04	56.57	94.44
FC-EF	65.60	55.01	57.65	52.20	93.58
MSCANet	93.95	90.16	93.91	87.73	97.49
SNUNet	**96.13**	91.52	93.89	87.11	97.36
RDPNet	94.76	92.79	94.32	90.09	97.84
DSIFN	91.30	89.08	89.13	85.34	96.70
DTCDSCN	89.74	85.24	87.09	80.24	96.56
BIT	93.82	89.70	90.08	86.12	97.07
ChangeFormer	94.40	92.48	93.94	89.80	97.74
Ours	95.48	**93.08**	**95.39**	**92.23**	**98.55**

To further demonstrate the effectiveness and superiority of our method, we also perform a quantitative analysis of the Season-Varying dataset. Table I shows the results of comparing the average numerical metrics of the various methods on the test dataset. As can be seen from Table I, our proposed method has optimal values in four evaluation metrics, Recall, F1, IoU, and OA, reaching 93.08%, 95.39%, 92.23%, and 98.55%, respectively. Compared with the best evaluation metrics of other methods, our method improves at least 1.07% and 2.14% in F1 and IoU metrics, respectively.

2) Evaluation Results on the LEVIR-CD Dataset: The visualization results obtained by our method and other methods on the LEVIR-CD dataset are shown in Fig. 14. It is worth noting

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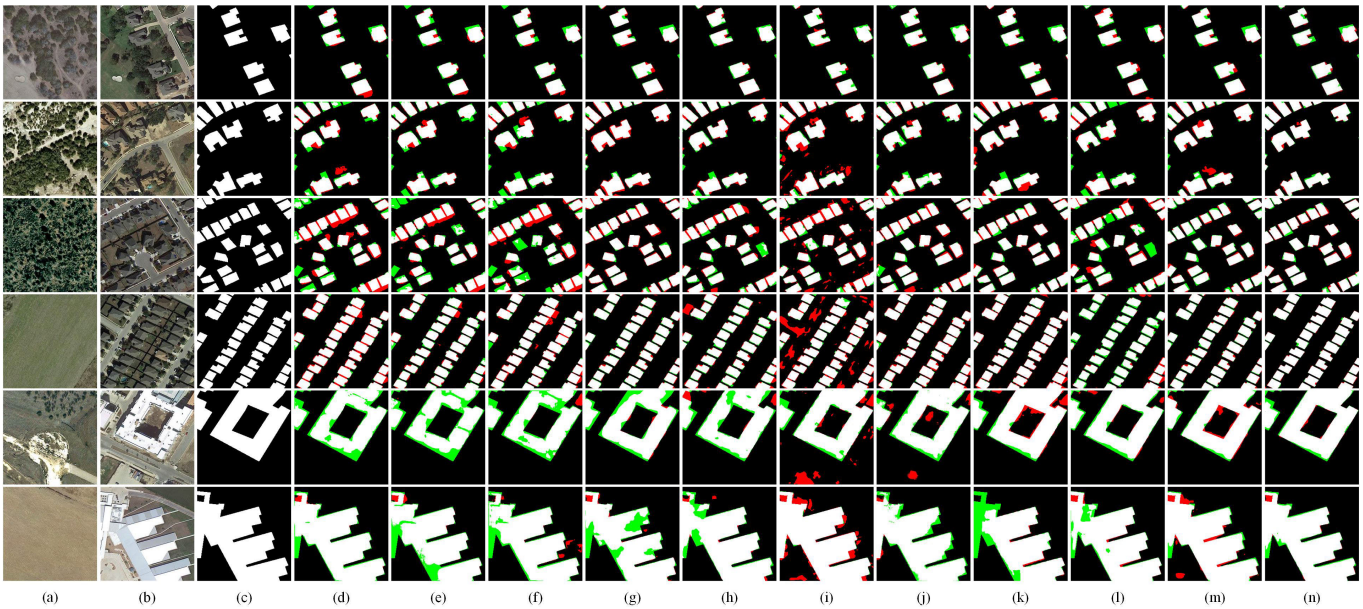


Fig. 14. Visualization results of different change detection methods on LEVIR-CD dataset. (a) and (b) Remote sensing images before and after the change. (c) Ground truth. (d) FC-Siam-Conc. (e) FC-Siam-Diff. (f) FC-EF. (g) SNUNet. (h) DTCDSCN. (i) DSIFN. (j) RDPNet. (k) MSCANet. (l) BIT. (m) ChangeFormer. (n) Ours. In the change map, white pixels represent TP, black pixels represent TN, red pixels represent FP, and green pixels represent FN.

that the LEVIR-CD dataset is a building change detection dataset. It focuses only on building changes and ignores other types of changes. The visual analysis in Fig. 14 shows that our method is very good at locating the correct area of change, whether it is a scattered or dense change of small buildings or a change of larger buildings.

Specifically, in the first two rows of images in Fig. 14, the changing buildings are more fragmented and there is irregular change. The prediction results of FC-Siam-Conc, DSIFN, MSCANet, and ChangeFormer have more false detections and poor detection of surrounding pseudochanges (trees and road changes). The predictions of FC-EF, DTCDSCN, and BIT did not detect all the change areas. Although FC-Siam-Diff, SNUNet, and RDPNet can detect the change part better, the detection of our method is better in comparison. In the images of the third and fourth rows of Fig. 14, the change buildings are densely distributed and small in size and are strongly disturbed by light shadows. From Fig. 14, it can be seen that there are more missed and false detections of FC-Siam-Diff, FC-EF, DSIFN, and BIT for small target area changes. The closer buildings in the prediction results of FC-Siam-Conc, FC-Siam-Diff, and FC-EF are directly connected together, and the clear object boundary cannot be segmented. Our method provides a clearer segmentation of the edges of changing targets while providing better differentiation between closely spaced buildings. In the images of the fifth and sixth rows of Fig. 14, the change in building volume is larger. The prediction results obtained by most algorithms are not clear and smooth enough in terms of the shape and boundary of the change target, and the missed detection area is large. On the contrary, our proposed algorithm learns the building features better, allowing the predicted change map to get a more complete boundary. The best overall results were achieved, although slightly missed detections occurred.

TABLE II

COMPARISON OF THE AVERAGE NUMERICAL INDICATORS OF DIFFERENT CHANGE DETECTION METHODS ON THE LEVIR-CD DATASET. VALUES ARE DESCRIBED USING PERCENTAGES, AND THE BEST VALUES ARE HIGHLIGHTED IN BOLD

Method	P (%)	R (%)	F1 (%)	IoU (%)	OA (%)
FC-Siam-Conc	89.65	84.85	83.32	72.46	97.59
FC-Siam-Diff	82.35	78.71	86.58	75.63	98.12
FC-EF	86.37	83.54	82.35	71.80	96.97
MSCANet	89.56	**91.20**	89.67	81.84	98.82
SNUNet	91.78	88.17	89.41	81.67	98.87
RDPNet	92.05	89.93	90.34	82.79	**98.96**
DSIFN	89.73	86.69	88.40	79.57	98.58
DTCDSCN	86.54	88.06	87.81	78.61	98.45
BIT	89.21	89.19	89.23	80.57	98.79
ChangeFormer	91.54	88.56	90.26	82.62	98.92
Ours	**92.72**	90.67	**91.48**	**83.39**	98.73

The quantitative results of the various methods on the LEVIR-CD dataset are given in Table II. As can be seen in Table II, our method achieves the best performance in several metrics compared with other methods, where the precision, F1, and IoU values are 92.72%, 91.48%, and 83.39%, respectively. Compared with the suboptimal method RDPNet, our method achieves performance improvements of 1.14% and 0.60% in F1 and IoU metrics, respectively.

3) Evaluation Results on the CLCD Dataset: The visualization results obtained by our method and other methods on the CLCD dataset are shown in Fig. 15. The CLCD dataset is a farmland change detection dataset with major change types, including buildings, roads, and bare land.

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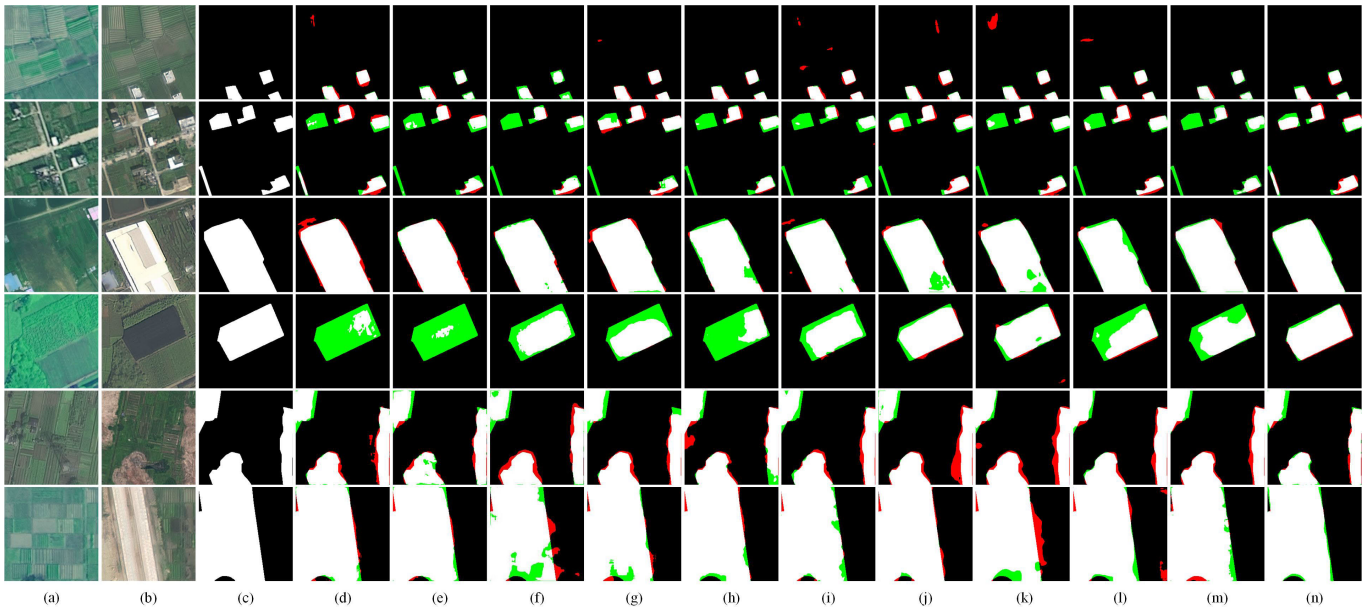


Fig. 15. Visualization results of different change detection methods on CLCD dataset. (a) and (b) Remote sensing images before and after the change. (c) Ground truth. (d) FC-Siam-Conc. (e) FC-Siam-Diff. (f) FC-EF. (g) SNUNet. (h) DTCDSCN. (i) DSIFN. (j) RDPNet. (k) MSCANet. (l) BIT. (m) ChangeFormer. (n) Ours. In the change map, white pixels represent TP, black pixels represent TN, red pixels represent FP, and green pixels represent FN.

The first three rows in Fig. 15 show the results of the change detection for different types of buildings. It can be seen that for the changes of small buildings (rows 1 and 2 in Fig. 15), the prediction results of most methods are more blurred at the edges of the change area due to the influence of the light angle, and there is a certain degree of false detection and omission. For the change of large buildings (row 3 in Fig. 15), the prediction results of FC-Siam-Conc and SNUNet have larger false detection areas. The rest of the methods can obtain the complete change area, but some detailed information is lost. Our method can not only effectively detect the small target change area (row 2 in Fig. 15) but also handle the edges of the large target change area better. The fourth and fifth rows in Fig. 15 show the results of change detection for bare land of different shapes. In the fourth row, except for RDPNet and MSCANet, which can identify the approximate change area, the prediction results of all other methods have different degrees of missed detection. In the fifth row, the irregular shape of the changed bare land makes it difficult to detect the contour of the changed area. Other methods either do not identify the complete area of change or have more redundant information. On the contrary, the prediction results of our method are significantly less in both false detection and missed detection areas, retain more information of the original image, and can better capture the contours of the changed areas. The sixth row in Fig. 15 shows the change detection results of the road. It can be seen that the prediction results of FC-EF, MSCANet, and BIT have more noise, and the remaining methods have a general outline description for the change area, but some details are missing in them. In contrast, our method has less redundant information and can accurately identify the target in the upper left corner. In conclusion, the visual analysis in Fig. 15 shows that our method predicts change maps with better detail and higher completeness of change areas.

TABLE III

COMPARISON OF THE AVERAGE NUMERICAL INDICATORS OF DIFFERENT CHANGE DETECTION METHODS ON THE CLCD DATASET. VALUES ARE DESCRIBED USING PERCENTAGES, AND THE BEST VALUES ARE HIGHLIGHTED IN BOLD

Method	P (%)	R (%)	F1 (%)	IoU (%)	OA (%)
FC-Siam-Conc	**73.97**	47.82	53.69	40.53	94.43
FC-Siam-Diff	71.19	42.61	50.02	37.92	94.39
FC-EF	67.23	37.05	48.36	34.07	94.32
MSCANet	67.51	**65.46**	60.04	42.59	94.81
SNUNet	65.40	56.17	60.82	43.63	94.90
RDPNet	71.35	58.34	62.53	45.66	95.04
DSIFN	64.18	50.11	59.79	41.98	94.65
DTCDSCN	62.80	48.49	57.47	40.81	94.59
BIT	67.05	51.49	59.93	42.12	94.77
ChangeFormer	69.36	53.64	61.31	44.29	94.98
Ours	72.44	62.70	**65.74**	**49.44**	**95.30**

Table III compares the performance of the various methods on the CLCD dataset. A quantitative analysis of Table III shows that our method achieves the best results for all three metrics, F1, IoU, and OA, reaching 65.74%, 49.44%, and 95.30%, respectively. Compared with the best evaluation metrics of other methods, our method improves F1 by at least 3.21% and IoU by 3.78%, respectively.

D. Model Efficiency Analysis

In addition to qualitative and quantitative analysis of the change detection results of all comparison methods, we also adopt three indexes, Params, floating-point operations per second (FLOPs), and Time, to further demonstrate the efficiency of the proposed network model. Params refer to the number

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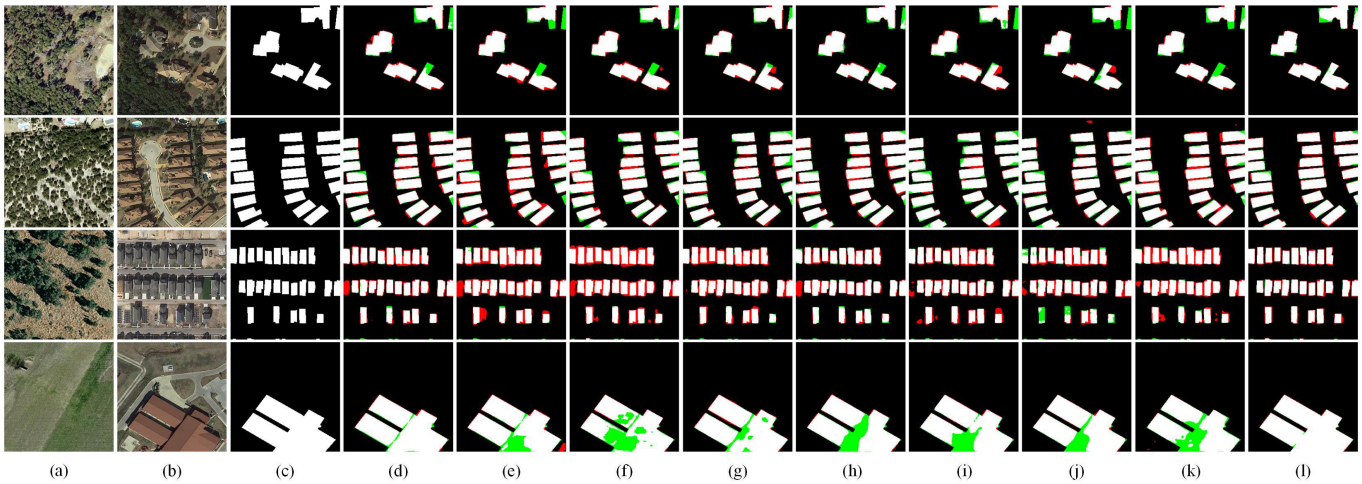


Fig. 16. Visualization results of the ablation experiment on the LEVIR-CD dataset. (a) and (b) Remote sensing images before and after the change. (c) Ground truth. (d) No FAPM. (e) No DFFM. (f) No CCT. (g) No CCA. (h) No TD. (i) No CEM. (j) No AAM. (k) No CCT + CCA. (l) Ours. In the change map, white pixels represent TP, black pixels represent TN, red pixels represent FP, and green pixels represent FN.

TABLE IV
COMPARISON OF THE EFFICIENCY OF DIFFERENT METHODS ON THE LEVIR-CD DATASET

Method	Params (M)	FLOPs (G)	Time (ms)
FC-Siam-Conc	1.55	5.32	0.725
FC-Siam-Diff	1.35	4.72	0.004
FC-EF	1.35	3.57	0.675
MSCANet	16.42	14.77	2.111
SNUNet	12.03	54.82	4.609
RDPNet	1.69	27.12	0.835
DSIFN	50.44	82.26	6.488
DTCDSCN	31.26	13.21	8.624
BIT	3.49	10.59	4.598
ChangeFormer	29.73	21.18	2.319
Ours	15.63	34.47	4.081

of parameters that need to be learned during model training, corresponding to the spatial complexity of the model. FLOPs refer to the number of floating point operations. The time complexity of the model is measured by calculating the sum of the number of multiplication operations and the number of addition operations. Time indicates the time that the model runs to process a single image.

Table IV gives the Params, FLOPs, and times for all methods. Among them, the smaller the Params and FLOPs, the less the complexity of the model. The smaller the time, the shorter the running time of the model. As can be seen from Table IV, our proposed network model is in the middle of the pack in terms of Params, FLOPs, and time metrics. However, by combining the previous subjective and objective analysis, we can know that our method gives the best results on different change detection datasets. Thus, our method can achieve a better balance in terms of model complexity, time cost, and accuracy. This also reflects the feasibility of our method in the change detection task from the side.

E. Ablation Study

To further validate the effectiveness of our proposed network model, we conduct ablation experiments on the LEVIR-CD dataset and the Season-Varying dataset to analyze the functions of different modules. Under the premise that the experimental environment, parameter settings, and image size are all uniform, we perform eight sets of ablation experiments. The quantitative evaluation results of each group of ablation experiments are shown in Table V. Not only that but Figs. 16 and 17 also provide the visual comparison of the ablation results on the two datasets.

Fully Attentional Pyramid Module: We perform an ablation of the FAPM to verify its effect on the experimental results. Our FAPM can extract and fuse features using a fully attentive approach and takes full account of the spatial location information in the feature map. As can be seen in Table V, when the FAPM is removed from the original network, the obtained metrics are reduced. For our main evaluation metrics F1 and IoU, the two datasets decrease by 3.23%/2.56% and 2.83%/3.19%, respectively. From Figs. 16 and 17, it can be seen that the model lacking FAPM can basically detect the change region, but there are still cases of false detection. This demonstrates that FAPM can improve the change detection performance of the network.

Dense Feature Fusion Module: We ablate the DFFM to verify its performance in the network. Specifically, we connect the extracted bitemporal features at different levels in the original network directly, without going through the DFFM module. The second row of Table V shows the evaluation results of removing DFFM. Comparison with the full model reveals that all metrics of removing DFFM from the network are significantly lower than those of the full model. Among them, for our main evaluation metrics F1 and IoU, the two datasets decrease by 5.24%/5.70% and 6.07%/6.44%, respectively. This may be due to the fact that DFFM can fully fuse bitemporal features and avoid feature misalignment. It can be seen from Figs. 16 and 17 that the change maps predicted by the model lacking DFFM have more misdetection and omission cases.

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TABLE V
RESULTS OF ABLATION EXPERIMENTS ON TWO CHANGE DETECTION DATASETS. VALUES ARE DESCRIBED USING PERCENTAGES, AND THE BEST VALUES ARE HIGHLIGHTED IN BOLD

Model	FAPM	DFFM	CCT	CCA	TD	CEM	AAM	LEVIR-CD Dataset					Season-Varying Dataset				
								P (%)	R (%)	F1 (%)	IoU (%)	OA (%)	P (%)	R (%)	F1 (%)	IoU (%)	OA (%)
AMCA	×	✓	✓	✓	✓	✓	✓	90.78	87.04	88.25	80.83	98.09	93.29	91.07	92.56	89.04	98.29
AMCA	✓	×	✓	✓	✓	✓	✓	89.53	83.92	86.24	77.69	97.36	89.43	83.61	89.32	85.79	97.26
AMCA	✓	✓	×	✓	✓	✓	✓	87.25	81.37	85.73	75.12	97.04	86.62	81.45	87.58	83.93	96.98
AMCA	✓	✓	✓	×	✓	✓	✓	91.76	89.43	89.92	82.27	98.35	94.37	92.12	93.93	90.60	98.34
AMCA	✓	✓	✓	✓	×	✓	✓	89.61	84.66	87.13	78.29	97.62	91.67	85.28	91.06	86.34	97.52
AMCA	✓	✓	✓	✓	✓	×	✓	90.22	85.86	87.70	78.90	97.94	91.18	89.94	92.19	87.90	97.85
AMCA	✓	✓	✓	✓	✓	✓	×	91.29	88.36	89.06	81.84	98.27	93.65	88.45	93.20	89.72	98.17
AMCA	✓	✓	×	×	✓	✓	✓	84.33	79.94	83.79	74.23	96.98	85.31	84.67	86.53	82.48	96.34
AMCA	✓	✓	✓	✓	✓	✓	✓	**92.72**	**90.67**	**91.48**	**83.39**	**98.73**	**95.48**	**93.08**	**95.39**	**92.23**	**98.55**

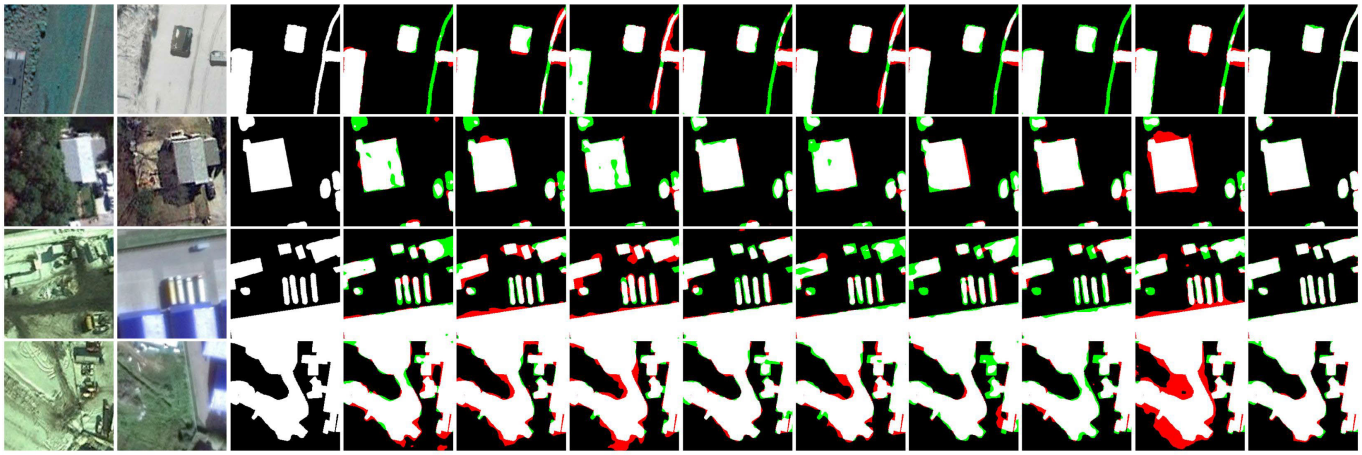


Fig. 17. Visualization results of the ablation experiment on the Season-Varying dataset. (a) and (b) Remote sensing images before and after the change. (c) Ground truth. (d) No FAPM. (e) No DFFM. (f) No CCT. (g) No CCA. (h) No TD. (i) No CEM. (j) No AAM. (k) No CCT + CCA. (l) Ours. In the change map, white pixels represent TP, black pixels represent TN, red pixels represent FP, and green pixels represent FN.

And the interference of pseudochange, such as trees and roads, is not well excluded, and the contour of the change area is not clear. This demonstrates the important contribution of DFFM to the change detection results.

Channelwise Cross Fusion Transformer and ChannelWise Cross Attention: To verify the effect of CCT and CCA on our network, we ablate them separately. First, we only remove the CCT from the network. Since our CCT can fuse channel features with different semantics and can learn spatiotemporal context information in the feature space. As shown in Table V, when the CCT is removed from the original network, the values obtained for each evaluation metric continue to decrease significantly. For our main evaluation metrics F1 and IoU, the two datasets decrease by 5.75%/8.27% and 7.81%/8.30%, respectively. Then, we only remove the CCA from the network. Our CCA enables the interaction of semantic information between different channels. As can be seen from the fourth row of data in Table V, the model lacking CCA decreases by 1.56%/1.12% and 1.46%/1.63% on the two datasets, respectively. Finally, we ablate CCT and CCA as a whole. As shown in the eighth row of Table V, the models lacking CCT and CCA decrease by 7.69%/9.16% and 8.86%/9.75% on the two datasets, respectively. It can be seen from Figs. 16 and 17 that the change map predicted by the model without CCT is more seriously disturbed by some nonbuildings. When the changing buildings are concentrated, there will be a sticky phenomenon, and the segmentation effect of the edge of the changing area is poor. The change map predicted by the model lacking CCA does not detect all the change areas, and there are some missed detections. The change maps predicted by the models lacking CCT and CCA have more false detections. This proves the essential role of both CCT and CCA in the overall performance improvement of the model.

Transformer Decoder: To demonstrate the effectiveness of our TD in the whole network, we ablate it. Specifically, we use a simple module instead of TD. As shown in Table V, for our main evaluation metrics F1 and IoU, the model lacking TD decreases by 4.35%/5.10% and 4.33%/5.89% on the two datasets, respectively. This may be because the cross-attention block provided in TD can project high-level semantic information into pixel space and generate finer feature maps. As can be seen in Figs. 16 and 17, the change map predicted by the model lacking TD loses too much of the original image information. And it is sensitive to the spectral features in the image that are not caused by the changing buildings, and the change detection is not effective. This shows that TD is an essential part of our network model.

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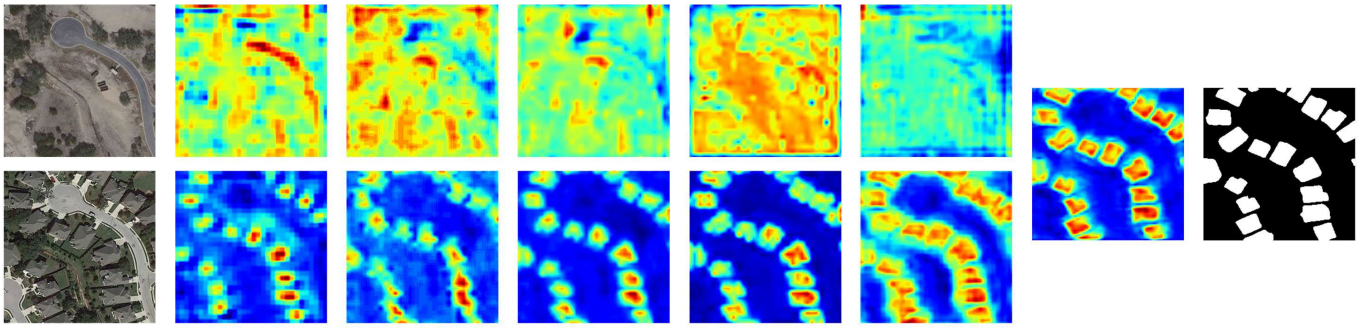


Fig. 18. Example of the visualization of the different stages in the network model. (a) Input bitemporal remote sensing image. (b) Deep features acquired by Siamese backbone network. (c) Feature map obtained by the FAPM. (d) Feature map output by CEM. (e) Feature map output by AAM. (f) Feature map refined by the TD. (g) Bitemporal feature difference map. (h) Predicted change map. The sample is from the LEVIR-CD dataset.

Context Extraction Module: We perform ablation experiments on the CEM to demonstrate its effect on the experimental results. Our CEM can effectively capture global and local features of feature maps and enhance feature representation. As shown in Table V, the network structure lacking the CEM is lower than the full model in all evaluation metrics. Among them, for our main evaluation metrics F1 and IoU, the two datasets decrease by 3.78%/4.49% and 3.20%/4.33%, respectively. As can be seen in Figs. 16 and 17, the change maps predicted by the model lacking CEM have more false detections and are less capable of refining the edges of the real change areas. This proves that CEM has an important role in our network.

Attention Aggregation Module: We ablate the AAM to verify its impact on network performance. Our AAM can fuse features from different scales together, improving the detection of details and edge information in images. As shown in Table V, for our main evaluation metrics F1 and IoU, the model lacking AAM decreases by 2.42%/1.55% and 2.19%/2.51% on the two datasets, respectively. As can be seen in Figs. 16 and 17, the change maps predicted by the model lacking AAM are poor for the detection of small target edges. This demonstrates the effectiveness of AAM in the overall network structure.

F. Network Visualization

To better understand the structure of our network, we visualize the features of the proposed method that are in several major stages, and the experimental results are shown in Fig. 18. Given the bitemporal remote sensing image [Fig. 18(a)], a Siamese feature extraction backbone generates a high-level feature mapping [Fig. 18(b)]. Then, the FAPM uses a fully attentional approach to extract and fuse deep features to produce a new feature map [Fig. 18(c)]. Fig. 18(d) shows the region of interest after the deep features are processed by CEM. Fig. 18(e) shows the feature map obtained by the AAM for the integration of low-level features and deep-level features. In addition, the high-level semantic information extracted by the channel transformer is projected into the pixel space using the TD to generate pixel-level features [Fig. 18(f)]. Finally, the feature difference map is calculated [Fig. 18(g)] and further processed by the prediction model to generate the predicted change map [Fig. 18(h)].

V. Conclusion

In this article, we propose an AMCA for remote sensing image change detection. In which, we use FAPM to further enhance the spatial location information in the feature map. And the deep features are extracted and fused by the full attentional approach. DFFM fully fuses the bitemporal features to reduce the effect of irrelevant changes on the detection results. Immediately after, CCT is introduced to effectively fuse together multiscale channel features focusing on different semantic patterns. CCA is then introduced to guide the channel information to eliminate the semantic gap between features at different scales. Meanwhile, we use an improved TD to map the acquired high-level semantic concepts into the original pixel space and optimize them, thus making the context information more comprehensive and obtaining pixel-level features. In addition, we use CEM to compute the local and global associations between semantic features and strengthen the ability of the network to extract detailed feature information. The addition of AAM can fully fuse the detailed features of low-level feature maps and the semantic features of deep-level feature maps and enhance the compactness between change pixels. Finally, extensive experiments on three change detection datasets demonstrate that our method is more effective than other methods in terms of both visual effects and evaluation metrics.

In addition, in future work, we will continue to investigate more effective methods for remote sensing image change detection to improve the robustness and accuracy of the network.

References

[1] M. Roy, S. Ghosh, and A. Ghosh, "A novel approach for change detection of remotely sensed images using semi-supervised multiple classifier system," *Inf. Sci.*, vol. 269, pp. 35–47, Jun. 2014.

[2] C. M. Viana, S. Oliveira, S. C. Oliveira, and J. Rocha, "Land use/land cover change detection and urban sprawl analysis," in *Spatial Modeling in GIS and R for Earth and Environmental Sciences*. Amsterdam, The Netherlands: Elsevier, 2019, pp. 621–651.

[3] R. Hedjam, A. Abdesselam, and F. Melgani, "Change detection from unlabeled remote sensing images using Siamese ANN," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2019, pp. 1530–1533.

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[4] P. Coppin, E. Lambin, I. Jonckheere, and B. Muys, "Digital change detection methods in natural ecosystem monitoring: A review," *Int. J. Remote Sens.*, vol. 25, no. 9, pp. 1565–1596, Sep. 2004.

[5] C. Wu, B. Du, X. Cui, and L. Zhang, "A post-classification change detection method based on iterative slow feature analysis and Bayesian soft fusion," *Remote Sens. Environ.*, vol. 199, pp. 241–255, Sep. 2017.

[6] G. Cao, Y. P. Li, Y. Liu, and Y. Shang, "Automatic change detection in high-resolution remote-sensing images by means of level set evolution and support vector machine classification," *Int. J. Remote Sens.*, vol. 35, no. 16, pp. 6255–6270, Aug. 2014.

[7] M. Volpi, D. Tuia, F. Bovolo, M. Kanevski, and L. Bruzzone, "Supervised change detection in VHR images using contextual information and support vector machines," *Int. J. Appl. Earth Observat. Geoinf.*, vol. 20, pp. 77–85, Feb. 2013.

[8] J. S. Deng, K. Wang, Y. H. Deng, and G. J. Qi, "PCA-based land-use change detection and analysis using multitemporal and multisensor satellite data," *Int. J. Remote Sens.*, vol. 29, no. 16, pp. 4823–4838, Aug. 2008.

[9] F. Bovolo and L. Bruzzone, "A theoretical framework for unsupervised change detection based on change vector analysis in the polar domain," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 1, pp. 218–236, Jan. 2006.

[10] C. Qiang and Y. Chen, "Multi-feature object-based change detection using self-adaptive weight change vector analysis," *Remote Sens.*, vol. 8, no. 7, p. 549, 2016.

[11] C. Wu, B. Du, and L. Zhang, "Slow feature analysis for change detection in multispectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 5, pp. 2858–2874, May 2014.

[12] C. Zhang, G. Li, and W. Cui, "High-resolution remote sensing image change detection by statistical-object-based method," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 7, pp. 2440–2447, Jul. 2018.

[13] J. L. Gil-Yepes, L. A. Ruiz, J. A. Recio, Á. Balaguer-Beser, and T. Hermosilla, "Description and validation of a new set of object-based temporal geostatistical features for land-use/land-cover change detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 121, pp. 77–91, Nov. 2016.

[14] L. Ma et al., "Object-based change detection in urban areas: The effects of segmentation strategy, scale, and feature space on unsupervised methods," *Remote Sens.*, vol. 8, no. 9, p. 761, 2016.

[15] Y. Zhang, D. Peng, and X. Huang, "Object-based change detection for VHR images based on multiscale uncertainty analysis," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 1, pp. 13–17, Jan. 2018.

[16] S. Toure et al., "An object-based temporal inversion approach to urban land use change analysis," *Remote Sens. Lett.*, vol. 7, no. 5, pp. 503–512, May 2016.

[17] L. Huang, G. Zhang, and Y. Li, "An object-based change detection approach by integrating intensity and texture differences," in *Proc. 2nd Int. Asia Conf. Informat. Control, Autom. Robot. (CAR)*, vol. 3, Mar. 2010, pp. 258–261.

[18] N. Wang, W. Li, R. Tao, and Q. Du, "Graph-based block-level urban change detection using Sentinel-2 time series," *Remote Sens. Environ.*, vol. 274, Jun. 2022, Art. no. 112993.

[19] X. Huang, D. Wen, J. Li, and R. Qin, "Multi-level monitoring of subtle urban changes for the megacities of China using high-resolution multi-view satellite imagery," *Remote Sens. Environ.*, vol. 196, pp. 56–75, Jul. 2017.

[20] C. Wu, L. Zhang, and B. Du, "Kernel slow feature analysis for scene change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 4, pp. 2367–2384, Jan. 2017.

[21] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.* Cham, Switzerland: Springer, 2015, pp. 234–241.

[22] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[23] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440.

[24] R. C. Daudt, B. Le Saux, and A. Boulch, "Fully convolutional Siamese networks for change detection," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 4063–4067.

[25] C. Zhang et al., "A deeply supervised image fusion network for change detection in high resolution bi-temporal remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 166, pp. 183–200, Aug. 2020.

[26] Y. Zhang, L. Fu, Y. Li, and Y. Zhang, "HDFNet: Hierarchical dynamic fusion network for change detection in optical aerial images," *Remote Sens.*, vol. 13, no. 8, p. 1440, Apr. 2021.

[27] S. Fang, K. Li, J. Shao, and Z. Li, "SNUNet-CD: A densely connected Siamese network for change detection of VHR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[28] H. Chen and Z. Shi, "A spatial–temporal attention-based method and a new dataset for remote sensing image change detection," *Remote Sens.*, vol. 12, no. 10, p. 1662, 2020.

[29] J. Chen et al., "DASNet: Dual attentive fully convolutional Siamese networks for change detection in high-resolution satellite images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 1194–1206, 2020.

[30] F. I. Diakogiannis, F. Waldner, and P. Caccetta, "Looking for change? Roll the dice and demand attention," *Remote Sens.*, vol. 13, no. 18, p. 3707, 2021.

[31] H. Jiang, X. Hu, K. Li, J. Zhang, J. Gong, and M. Zhang, "PGA-SiamNet: Pyramid feature-based attention-guided Siamese network for remote sensing orthoimagery building change detection," *Remote Sens.*, vol. 12, no. 3, p. 484, Feb. 2020.

[32] X. Peng, R. Zhong, Z. Li, and Q. Li, "Optical remote sensing image change detection based on attention mechanism and image difference," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7296–7307, Sep. 2021.

[33] J. Xu, C. Luo, X. Chen, S. Wei, and Y. Luo, "Remote sensing change detection based on multidirectional adaptive feature fusion and perceptual similarity," *Remote Sens.*, vol. 13, no. 15, p. 3053, Aug. 2021.

[34] M. Zhang and W. Shi, "A feature difference convolutional neural network-based change detection method," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 10, pp. 7232–7246, Oct. 2020.

[35] M. Zhang, G. Xu, K. Chen, M. Yan, and X. Sun, "Triplet-based semantic relation learning for aerial remote sensing image change detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 2, pp. 266–270, Feb. 2019.

[36] H. Wang, P. Cao, J. Wang, and O. R. Zaiane, "UCTransNet: Rethinking the skip connections in U-Net from a channel-wise perspective with transformer," in *Proc. AAAI Conf. Artif. Intell.*, 2022, vol. 36, no. 3, pp. 2441–2449.

[37] Y. Zhang, W. Li, M. Zhang, Y. Qu, R. Tao, and H. Qi, "Topological structure and semantic information transfer network for cross-scene hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Sep. 16, 2021, doi: 10.1109/TNNLS.2021.3109872.

[38] Y. Zhang, W. Li, R. Tao, J. Peng, Q. Du, and Z. Cai, "Cross-scene hyperspectral image classification with discriminative cooperative alignment," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 9646–9660, Nov. 2021.

[39] Y. Zhang, W. Li, M. Zhang, S. Wang, R. Tao, and Q. Du, "Graph information aggregation cross-domain few-shot learning for hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Jun. 30, 2022, doi: 10.1109/TNNLS.2022.3185795.

[40] Y. Zhang, W. Li, W. Sun, R. Tao, and Q. Du, "Single-source domain expansion network for cross-scene hyperspectral image classification," 2022, *arXiv:2209.01634*.

[41] Y. Zhang, M. Zhang, W. Li, S. Wang, and R. Tao, "Language-aware domain generalization network for cross-scene hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–12, 2023.

[42] J. Li, X. Feng, and Z. Hua, "Low-light image enhancement via progressive-recursive network," *IEEE Trans. Circuits Syst. Video Techn.*, vol. 31, no. 11, pp. 4227–4240, Nov. 2021.

[43] Z. Li, J. Li, F. Zhang, and L. Fan, "CADUI: Cross-attention-based depth unfolding iteration network for pansharpening remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–20, 2023.

[44] X. Niu, M. Gong, T. Zhan, and Y. Yang, "A conditional adversarial network for change detection in heterogeneous images," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 1, pp. 45–49, Jan. 2018.

[45] D. Peng, Y. Zhang, and H. Guan, "End-to-end change detection for high resolution satellite images using improved UNet++," *Remote Sens.*, vol. 11, no. 11, p. 1382, 2019.

[46] R. C. Daudt, B. Le Saux, A. Boulch, and Y. Gousseau, "Multitask learning for large-scale semantic change detection," *Comput. Vis. Image Understand.*, vol. 187, Oct. 2019, Art. no. 102783.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:53:33 UTC from IEEE Xplore. Restrictions apply.

[47] Y. Liu, C. Pang, Z. Zhan, X. Zhang, and X. Yang, "Building change detection for remote sensing images using a dual-task constrained deep Siamese convolutional network model," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 5, pp. 811–815, May 2021.

[48] H. Chen, C. Wu, B. Du, L. Zhang, and L. Wang, "Change detection in multisource VHR images via deep Siamese convolutional multiple-layers recurrent neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 4, pp. 2848–2864, Apr. 2020.

[49] K. Li, Z. Li, and S. Fang, "Siamese NestedUNet networks for change detection of high resolution satellite image," in *Proc. 1st Int. Conf. Control, Robot. Intell. Syst.*, Oct. 2020, pp. 42–48.

[50] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7132–7141.

[51] H. Touvron, M. Cord, M. Douze, F. Massa, A. Sablayrolles, and H. Jégou, "Training data-efficient image transformers & distillation through attention," in *Proc. Int. Conf. Mach. Learn.*, 2021, pp. 10347–10357.

[52] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[53] Z. Dai, B. Cai, Y. Lin, and J. Chen, "UP-DETR: Unsupervised pre-training for object detection with transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 1601–1610.

[54] Z. Sun, S. Cao, Y. Yang, and K. Kitani, "Rethinking transformer-based set prediction for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 3611–3620.

[55] S. Zheng et al., "Rethinking semantic segmentation from a sequence-to-sequence perspective with transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 6881–6890.

[56] B. Wu et al., "Visual transformers: Token-based image representation and processing for computer vision," 2020, *arXiv:2006.03677*.

[57] F. Yang, H. Yang, J. Fu, H. Lu, and B. Guo, "Learning texture transformer network for image super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 5791–5800.

[58] H. Chen et al., "Pre-trained image processing transformer," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 12299–12310.

[59] X. Su, J. Li, and Z. Hua, "Transformer-based regression network for pansharpening remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–23, 2022.

[60] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022.

[61] Q. Li, R. Zhong, X. Du, and Y. Du, "TransUNetCD: A hybrid transformer network for change detection in optical remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–19, 2022.

[62] W. G. C. Bandara and V. M. Patel, "A transformer-based Siamese network for change detection," 2022, *arXiv:2201.01293*.

[63] Y. Feng, H. Xu, J. Jiang, H. Liu, and J. Zheng, "ICIF-Net: Intra-scale cross-interaction and inter-scale feature fusion network for bitemporal remote sensing images change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2022.

[64] C. Zhang, L. Wang, S. Cheng, and Y. Li, "SwinSUNet: Pure transformer network for remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2022.

[65] G. Wang, B. Li, T. Zhang, and S. Zhang, "A network combining a transformer and a convolutional neural network for remote sensing image change detection," *Remote Sens.*, vol. 14, no. 9, p. 2228, May 2022.

[66] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2021, pp. 10012–10022.

[67] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 3–19.

[68] W. Wang, X. Tan, P. Zhang, and X. Wang, "A CBAM based multiscale transformer fusion approach for remote sensing image change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 6817–6825, 2022.

[69] Q. Wang, B. Wu, P. Zhu, P. Li, W. Zuo, and Q. Hu, "ECA-Net: Efficient channel attention for deep convolutional neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 11534–11542.

[70] X. Wang, R. Girshick, A. Gupta, and K. He, "Non-local neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7794–7803.

[71] J. Fu et al., "Dual attention network for scene segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 3146–3154.

[72] L. Mi and Z. Chen, "Superpixel-enhanced deep neural forest for remote sensing image semantic segmentation," *ISPRS J. Photogramm. Remote Sens.*, vol. 159, pp. 140–152, 2020.

[73] Q. Song, J. Li, C. Li, H. Guo, and R. Huang, "Fully attentional network for semantic segmentation," in *Proc. AAAI Conf. Artif. Intell.*, 2022, vol. 36, no. 2, pp. 2280–2288.

[74] M. Lebedev, Y. V. Vizilter, O. Vygolov, V. Knyaz, and A. Y. Rubis, "Change detection in remote sensing images using conditional adversarial networks," *Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 42, no. 2, pp. 565–571, May 2018.

[75] M. Liu, Z. Chai, H. Deng, and R. Liu, "A CNN-transformer network with multiscale context aggregation for fine-grained cropland change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 4297–4306, 2022.

[76] H. Chen, F. Pu, R. Yang, R. Tang, and X. Xu, "RDP-Net: Region detail preserving network for change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–10, 2022.



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